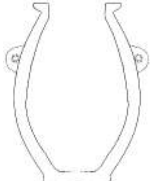
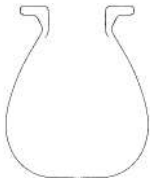
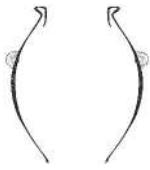
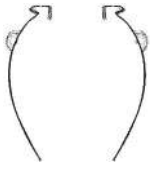
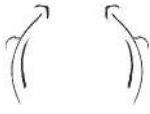
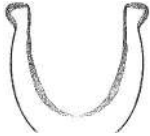
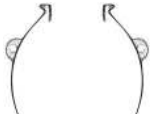
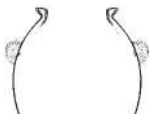
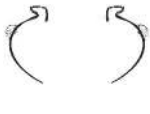
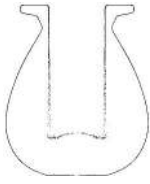
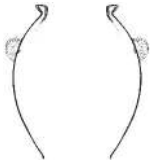
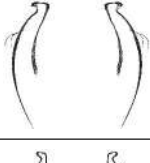
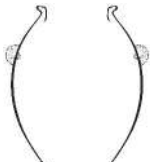
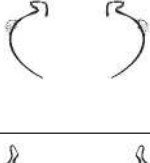
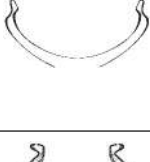


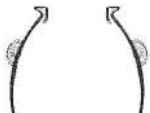
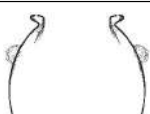
Artifact Catalogue - Ranked By Precision Score

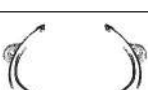
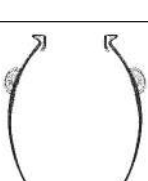
Find detailed description of the precision score metric, page 7.

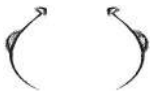
Artifact			Precision Score	Collection	References
Artifact image pending		REF01	43 943		Report Publication
		General shape:	Full: 43 943 Exterior: 43 943		Precision overview, see page 61
		PV001	1980	In private collection of Adam Young	Report Publication
		General shape: Red Granite	Full: 1177 Exterior: 1980 Interior: 798		Precision overview, see page 8
		MV003	24	Petrie Museum of Egyptian Archaeology	Report Publication
		General shape: Andesite Porphyry	Full: 25 Exterior: 24 Interior: 748		Precision overview, see page 15
		PV006	621	In private collection	Report Publication
		General shape: Dark grey granite	Full: 610 Exterior: 621 Interior: 479		Precision overview, see page 9
		MV020	9	Petrie Museum of Egyptian Archaeology	Report Publication
		General shape: Basalt	Full: 9 Exterior: 9 Interior: 464		Precision overview, see page 32
		MV017	30	Petrie Museum of Egyptian Archaeology	Report Publication
		General shape: Limestone	Full: 30 Exterior: 30 Interior: 342		Precision overview, see page 29
		MV026	39	Petrie Museum of Egyptian Archaeology	Report Publication
		General shape: Basalt	Full: 59 Exterior: 39 Interior: 322		Precision overview, see page 53
		MV006	11	Petrie Museum of Egyptian Archaeology	Report Publication
		General shape: Basalt	Full: 16 Exterior: 11 Interior: 274		Precision overview, see page 18
		MV014	4	Petrie Museum of Egyptian Archaeology	Report Publication
		General shape: Basalt	Full: 86 Exterior: 54 Interior: 243		Precision overview, see page 26
		MV027	7	Petrie Museum of Egyptian Archaeology	Report Publication
		General shape: Basalt	Full: 13 Exterior: 7 Interior: 234		Precision overview, see page 54

		MV027b General shape: Basalt	8 Full: 12 Exterior: 8 Interior: 223	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 55
		MV008 General shape: Diorite	200 Full: 100 Exterior: 200	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 20
		MV020b General shape: Basalt	7 Full: 10 Exterior: 7 Interior: 180	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 43
		MV005 General shape: Limestone	114 Full: 114 Exterior: 114 Interior: 134	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 17
		MV003b General shape: Andesite Porphyry	19 Full: 21 Exterior: 19 Interior: 124	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 34
		MV004 General shape: Basalt	72 Full: 73 Exterior: 72 Interior: 109	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 16
		MV020c General shape: Basalt	8 Full: 10 Exterior: 8 Interior: 97	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 44
		MV021 General shape: Grey Basalt	10 Full: 10 Exterior: 10 Interior: 91	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 33
		MV030 General shape: Basalt	26 Full: 39 Exterior: 26 Interior: 87	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 58
		MV018 General shape: Diorite	13 Full: 14 Exterior: 13 Interior: 86	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 30

		RV001 General shape: Red Granite	84 Full: 24 Exterior: 84 Interior: 4	In private collection	Report Publication Precision overview, see page 10
		MV004b General shape: Basalt	70 Full: 73 Exterior: 70 Interior: 83	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 35
		MV004c General shape: Basalt	70 Full: 72 Exterior: 70 Interior: 79	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 36
		MV028 General shape: Basalt	10 Full: 12 Exterior: 10 Interior: 75	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 56
		MV021b General shape: Grey Basalt	9 Full: 12 Exterior: 9 Interior: 66	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 45
		MV018b General shape: Diorite	13 Full: 19 Exterior: 13 Interior: 65	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 41
		MV009 General shape: Diorite	15 Full: 21 Exterior: 15 Interior: 63	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 21
		MV015c General shape: Black & White Syenite, perhaps Hornblende Diorite	52 Full: 31 Exterior: 52 Interior: 12	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 38
		MV015b General shape: Black & White Syenite, perhaps Hornblende Diorite	50 Full: 33 Exterior: 50 Interior: 14	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 37
		RV002 General shape: Diorite	47 Full: 27 Exterior: 47 Interior: 7	In private collection	Report Publication Precision overview, see page 11

		MV011 General shape: Porphyry	1.81 Full: 1.87 Exterior: 1.81 Interior: 40	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 23
		MV023 General shape: Basalt	21 Full: 23 Exterior: 21 Interior: 39	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 47
Artifact image pending		MV032 General shape: Basalt	17 Full: 21 Exterior: 17 Interior: 38	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 60
		MV022 General shape: Basalt	7 Full: 9 Exterior: 7 Interior: 37	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 46
		MV012 General shape: Basalt	6 Full: 8 Exterior: 6 Interior: 33	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 24
		MV029 General shape: Basalt	6 Full: 8 Exterior: 6 Interior: 32	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 57
		MV019b General shape: Porphyry	15 Full: 17 Exterior: 15 Interior: 31	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 42
		MV023b General shape: Basalt	17 Full: 18 Exterior: 17 Interior: 28	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 48
		MV016 General shape: Basalt	2.22 Full: 2.29 Exterior: 2.22 Interior: 28	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 28
		MV015 General shape: Black & White Syenite, perhaps	26 Full: 24 Exterior: 26 Interior: 9	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 27

Hornblende Diorite					
		MV019 General shape: Porphyry	14 Full: 15 Exterior: 14 Interior: 21	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 31
		MV007 General shape: Granite	19 Full: 18 Exterior: 19	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 19
		MV024b General shape: Basalt	1.86 Full: 4 Exterior: 1.86 Interior: 18	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 50
		MV031 General shape: Basalt	8 Full: 10 Exterior: 8 Interior: 16	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 59
		MV024 General shape: Basalt	1.84 Full: 4 Exterior: 1.84 Interior: 14	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 49
		MV025 General shape: Basalt	4 Full: 7 Exterior: 4 Interior: 14	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 51
		MV025b General shape: Basalt	3 Full: 5 Exterior: 3 Interior: 13	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 52
		MV016b General shape: Basalt	2.12 Full: 2.83 Exterior: 2.12 Interior: 12	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 39
		MV010 General shape: Calcite (Egyptian Alabaster)	1.17 Full: 1.32 Exterior: 1.17 Interior: 11	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 22
		MV001 General shape: Pottery	1.93 Full: 2.01 Exterior: 1.93 Interior: 10	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 13

		MV013 General shape: Diorite	4 Full: 5 Exterior: 4 Interior: 6	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 25
		MV016c General shape: Basalt	2.27 Full: 2.49 Exterior: 2.27 Interior: 2.76	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 40
		RV003 General shape: Marble breccia	1.46 Full: 1.49 Exterior: 1.46 Interior: 1.53	In private collection	Report Publication Precision overview, see page 12
		MV002 General shape: Breccia	1.02 Full: 0.94 Exterior: 1.02	Petrie Museum of Egyptian Archaeology	Report Publication Precision overview, see page 14

Precision Score Of The Artifact

To enable valid comparison of the manufacturing precision of different artifacts, a metric that robustly quantifies the overall precision of the object is required. The considerations for such a metric will be explored in this section.

Based on these considerations, a *Precision Score* metric will be defined.

For an object to be described as having been manufactured with high precision, several qualities must be present *concurrently*, and throughout the *entire* geometry of the final object. A given object may exhibit high levels of one or more *components* of precision, but be lacking in others. For example:

- An object may present high levels of coaxiality, but lack circularity.
- An object may exhibit good circularity, but show imperfections in the surface structure.
- An object may be smoothed to perfection *without* any circularity or coaxiality.
- An object may exhibit high levels of all of the above metrics in *some* areas, but not in others.

Therefore, a precision score metric **must** account for *all* aspects of the individual, underlying precision metrics (circularity, concentricity, coaxiality and surface variability) throughout the *entire* surface area of the object.

The composite high order polynomial model, used to generate the surface variability map (detailed in the full report) is the best continuous mathematical representation of the object available to us (lacking any original design plans, as would normally be available in metrological analysis). This idealized model encompasses all of the above component metrics.

In the creation of the model, all scan data-points are taken into account (excluding areas with extensive damage), making it the best possible idealized representation we can achieve. When this model has been accurately created, the deviation between the model and the scanned data-points can be calculated over the non-discretized polynomials, *without* the need for an “original” CAD model (and importantly, unless such a CAD model *actually* corresponded to the original design intent, it would be an insufficient comparison basis).

Within the context of defining a valid, overall precision metric, this approach satisfies the incorporation of all of the necessary metrics:

- **Circularity:** Because the reconstructed polynomial model is revolved around the Z-plane, the idealized representation is perfectly circular, and thus incorporates the circularity component.
- **Concentricity and coaxiality:** Because the Z-axis (datum axis) is the center axis of the model, it incorporates the concentricity and coaxiality components.
- **Surface variability:** Because the model is continuous and non-discretized, it can be used accurately for all points of the scan data, and incorporates the surface variability component.

The level of precision ultimately achieved in a physical object does not share a linear relationship with its manufacturing requirements. Since continuously higher levels of final precision becomes progressively harder to achieve, an overall precision metric must take this relationship into account.

A robust statistical metric that satisfies this requirement is the *Mean Squared Deviation* (MSD or MSE). Here specifically, we can utilize the mean square of the deviations between the model ($\hat{y}$) and the data-points (y_i).

Combining all of the above considerations, we can express a well-defined *Precision Score* metric, that provides an immediately accessible way to understand the overall precision of an object, while being statistically valid. Since the Mean Squared Deviation tends towards zero as the overall precision increases, the inverse of the Mean Squared Deviation is taken to obtain a precision score metric that increases as precision increases¹:

$$\text{Precision Score} = \frac{n}{\sum_{i=1}^n (y_i - \hat{y})^2}$$

A precision score will be calculated separately for:

- The exterior surface
- The interior surface
- The full surface

As most scans do not include sufficient scan data for the interior surface, the exterior surface will be used for calculating the precision score in most cases. In the rare case that the scan data is more complete for the *interior* surface, this will be used instead.

¹The precision score unit is $\frac{1}{\text{mm}^2}$

PV001



Artifact Foundation



Artifact Precision Report, version 01.20	
Material	Red Granite
Description	Egyptian Early Dynastic Granite Amphora Jar, 2nd-3rd Dynasty, c. 2900-2600 B.C.
Collection	In private collection of Adam Young
Provenance	Previously in collection of Mr. David Lewis, London UK, acquired in 1980s
Precision Score	Exterior 1980, Interior 798

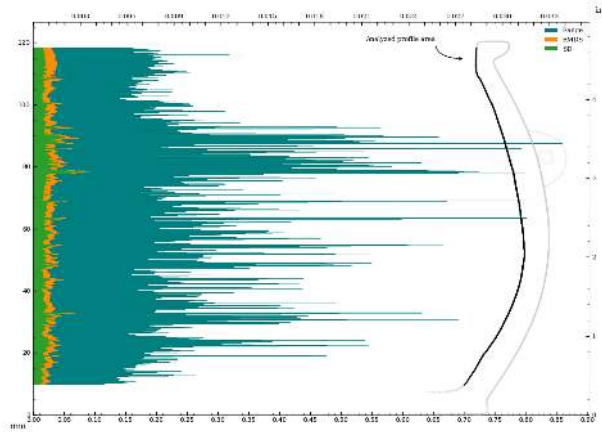
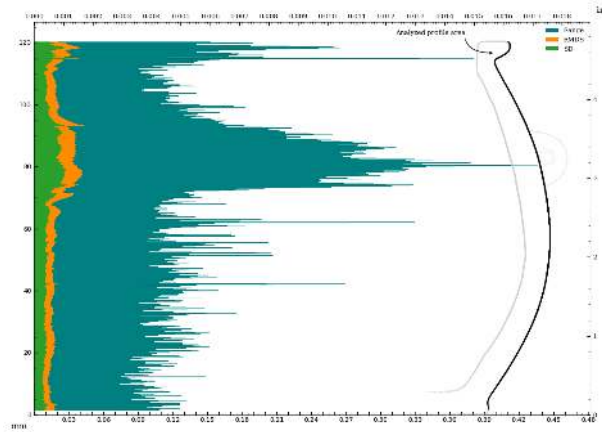
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

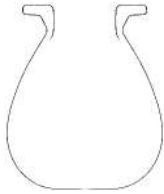
Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0005	0.022	0.015	0.008	0.637	−0.365	0.272	7267732
Interior	0.0014	0.037	0.025	0.013	1.093	−0.319	0.775	6200241
Interior separate	0.0013	0.035	0.024	0.012	1.082	−0.321	0.761	6200250
	in ²	in	in	in	in	in	in	
Exterior	0.000001	0.0009	0.0006	0.0003	0.0251	−0.0144	0.0107	7267732
Interior	0.000002	0.0015	0.0010	0.0005	0.0430	−0.0125	0.0305	6200241
Interior separate	0.000002	0.0014	0.0010	0.0005	0.0426	−0.0126	0.0300	6200250

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.123	0.070	0.459	0.011	0.006	0.035	0.018	0.011	0.043	2377	0.050
Interior	0.234	0.109	0.860	0.020	0.011	0.080	0.033	0.018	0.092	2179	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0049	0.0028	0.0181	0.0004	0.0003	0.0014	0.0007	0.0004	0.0017	2377	0.0020
Interior	0.0092	0.0043	0.0339	0.0008	0.0004	0.0031	0.0013	0.0007	0.0036	2179	0.0020



PV006



Artifact Precision Report, version 01.20

Material Dark grey granite

Description Artifact in private collection.

Collection In private collection

Provenance Unknown

Precision Score Exterior 621, Interior 479

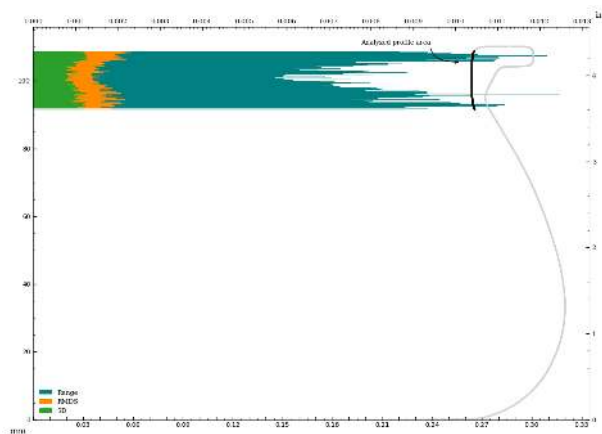
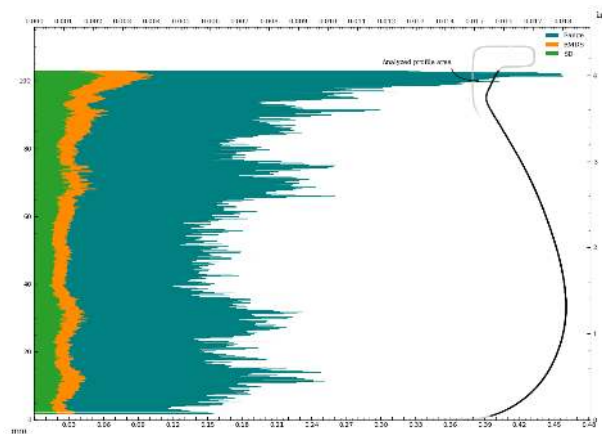
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

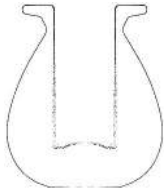
Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0016	0.040	0.026	0.015	0.485	−0.247	0.238	511432
Interior	0.0066	0.081	0.046	0.033	0.461	−0.264	0.197	34262
Interior separate	0.0021	0.046	0.027	0.018	0.372	−0.213	0.159	34261
	in ²	in	in	in	in	in	in	
Exterior	0.000002	0.0016	0.0010	0.0006	0.0191	−0.0097	0.0094	511432
Interior	0.000010	0.0032	0.0018	0.0013	0.0182	−0.0104	0.0078	34262
Interior separate	0.000003	0.0018	0.0011	0.0007	0.0147	−0.0084	0.0063	34261

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.182	0.111	0.458	0.021	0.013	0.062	0.036	0.024	0.108	1014	0.100
Interior	0.213	0.139	0.319	0.026	0.018	0.037	0.044	0.033	0.063	172	0.100
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0072	0.0044	0.0180	0.0008	0.0005	0.0024	0.0014	0.0009	0.0043	1014	0.0039
Interior	0.0084	0.0055	0.0125	0.0010	0.0007	0.0015	0.0017	0.0013	0.0025	172	0.0039



RV001



Artifact Precision Report, version 01.20

Material

Red Granite

Description

Modern CNC machined replica of vessel PV006, dimensions have been altered.

Collection

In private collection

Provenance

NA

Precision Score

Exterior 84, Interior 4

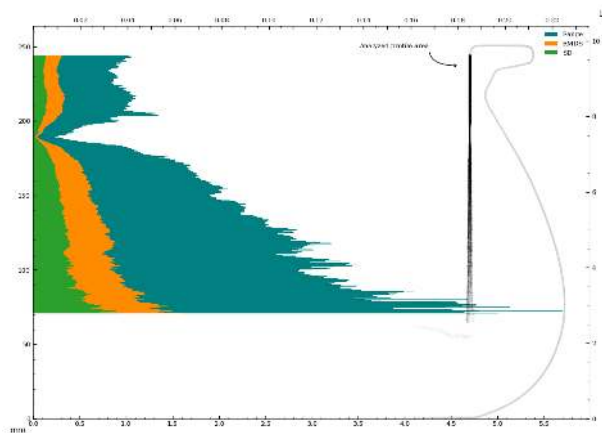
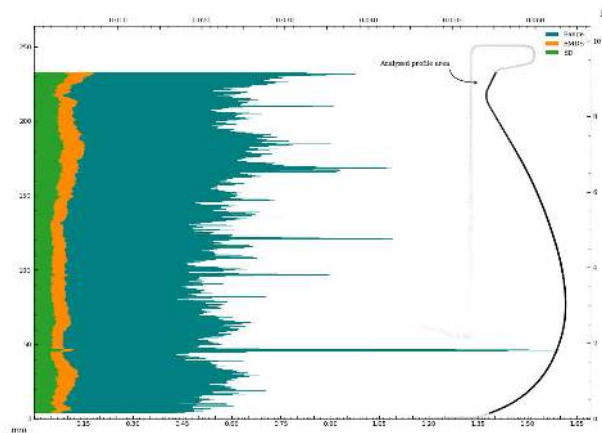
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0119	0.109	0.066	0.043	1.800	−1.374	0.425	7140686
Interior	0.4133	0.643	0.385	0.210	8.951	−2.141	6.810	914255
Interior separate	0.2718	0.521	0.339	0.175	8.347	−1.890	6.457	914236
	in ²	in	in	in	in	in	in	
Exterior	0.000018	0.0043	0.0026	0.0017	0.0709	−0.0541	0.0167	7140686
Interior	0.000641	0.0253	0.0151	0.0083	0.3524	−0.0843	0.2681	914255
Interior separate	0.000421	0.0205	0.0133	0.0069	0.3286	−0.0744	0.2542	914236

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		
Exterior	0.587	0.428	1.610	0.062	0.048	0.106	0.106	0.079	0.182	2294	0.100
Interior	1.915	0.228	5.702	0.277	0.024	0.963	0.593	0.043	1.626	1735	0.100
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0231	0.0168	0.0634	0.0025	0.0019	0.0042	0.0042	0.0031	0.0072	2294	0.0039
Interior	0.0754	0.0090	0.2245	0.0109	0.0010	0.0379	0.0233	0.0017	0.0640	1735	0.0039



RV002



Scientists Against Myths



Artifact Precision Report, version 01.20

Material Diorite

Description Modern replica - Produced by Olga Vdovina and Yulia Gukasova in collaboration with Scientists Against Myths

Collection In private collection

Provenance NA

Precision Score Exterior 47, Interior 7

Scan note Mesh density is very low compared to other vessel scans

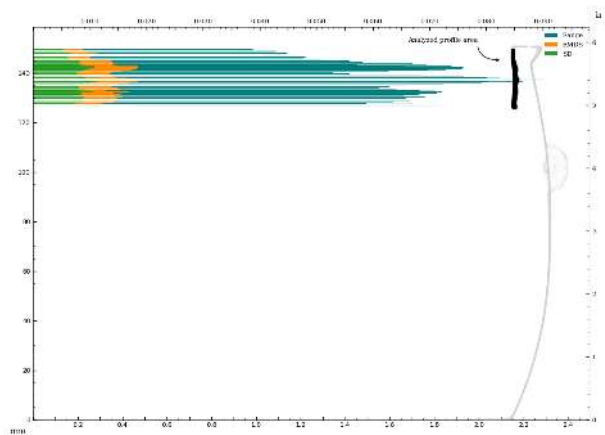
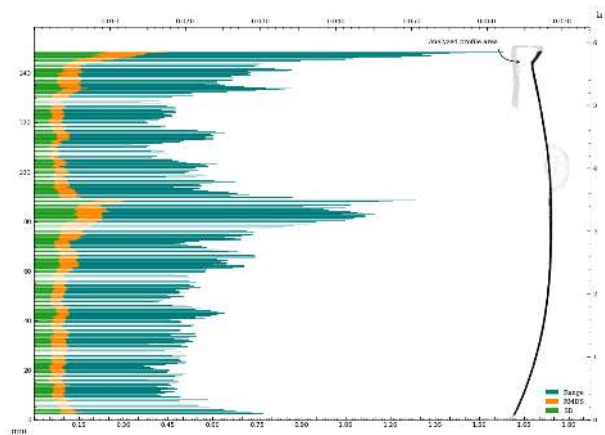
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0214	0.146	0.097	0.053	1.668	−1.082	0.586	184585
Interior	0.4486	0.670	0.347	0.254	3.305	−1.364	1.940	27793
Interior separate	0.1381	0.372	0.228	0.147	2.536	−0.968	1.568	27815
	in ²	in	in	in	in	in	in	
Exterior	0.000033	0.0058	0.0038	0.0021	0.0657	−0.0426	0.0231	184585
Interior	0.000695	0.0264	0.0137	0.0100	0.1301	−0.0537	0.0764	27793
Interior separate	0.000214	0.0146	0.0090	0.0058	0.0999	−0.0381	0.0617	27815

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.532	0.302	1.784	0.066	0.043	0.268	0.113	0.066	0.447	980	0.150
Interior	1.677	0.920	2.376	0.214	0.115	0.382	0.368	0.215	0.536	155	0.150
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0210	0.0119	0.0702	0.0026	0.0017	0.0105	0.0044	0.0026	0.0176	980	0.0059
Interior	0.0660	0.0362	0.0935	0.0084	0.0045	0.0150	0.0145	0.0085	0.0211	155	0.0059



RV003



Scientists Against
Myths



Artifact Precision Report, version 01.20

Material	Marble breccia
Description	Modern replica - Produced by Olga Vdovina and Yulia Gukasova in collaboration with Scientists Against Myths
Collection	In private collection
Provenance	NA
Precision Score	Exterior 1.46, Interior 1.53

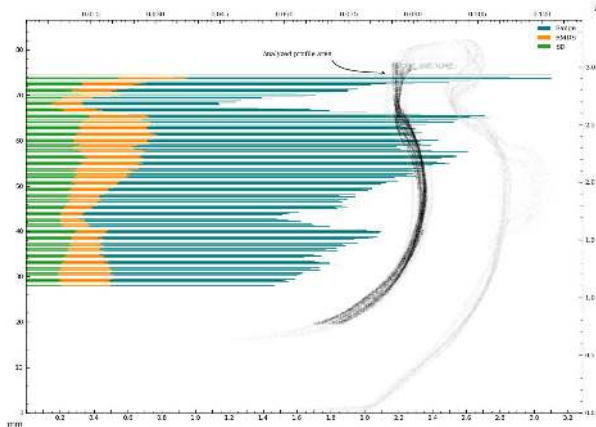
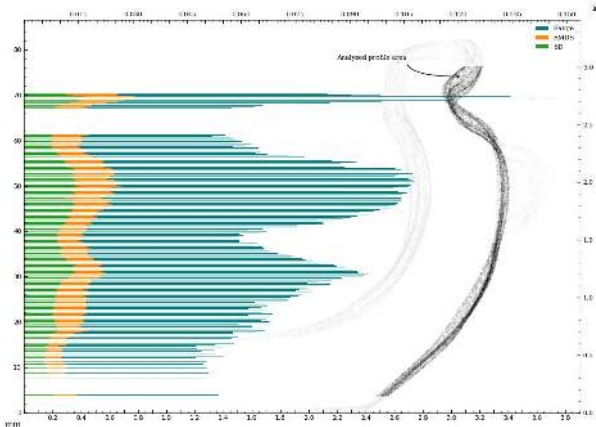
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.6847	0.827	0.538	0.299	5.346	−2.709	2.638	108882
Interior	1.8634	1.365	0.677	0.514	6.116	−3.310	2.806	91258
Interior separate	0.6520	0.807	0.566	0.253	10.710	−2.854	7.856	93028
	in ²	in	in	in	in	in	in	
Exterior	0.001061	0.0326	0.0212	0.0118	0.2105	−0.1066	0.1038	108882
Interior	0.002888	0.0537	0.0267	0.0202	0.2408	−0.1303	0.1105	91258
Interior separate	0.001011	0.0318	0.0223	0.0099	0.4216	−0.1124	0.3093	93028

Circularity measurements

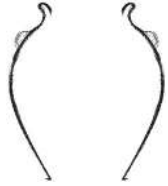
Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.913	1.205	3.721	0.251	0.140	0.489	0.457	0.235	0.916	269	0.200
Interior	1.962	1.070	3.129	0.266	0.118	0.603	0.501	0.314	1.057	235	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0753	0.0474	0.1465	0.0099	0.0055	0.0193	0.0180	0.0092	0.0361	269	0.0079
Interior	0.0772	0.0421	0.1232	0.0105	0.0046	0.0237	0.0197	0.0123	0.0416	235	0.0079



MV001



Petrie Museum, CC BY-NC-SA



Artifact Precision Report, version 01.20	
Material	Pottery
Description	Decorated pot, type D 2 k; wavy handled. From Hu tomb U 78.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology
Precision Score	Exterior 1.93, Interior 10

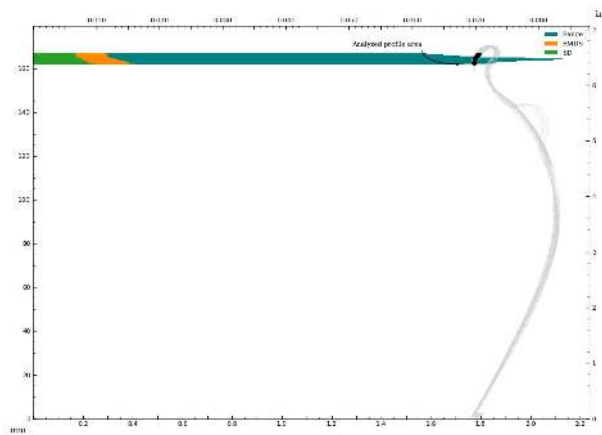
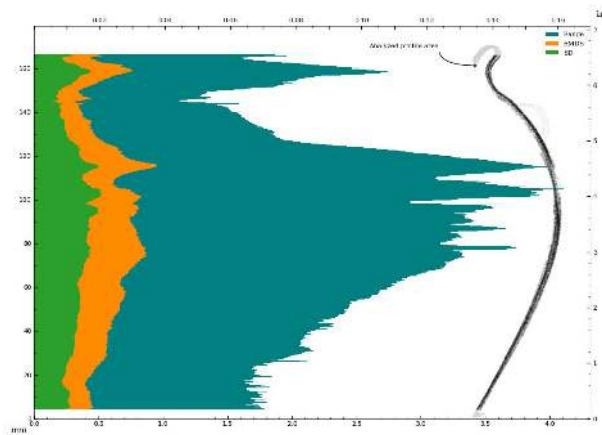
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.5193	0.721	0.424	0.299	6.565	−3.374	3.190	12132817
Interior	0.5405	0.735	0.348	0.278	3.610	−1.700	1.910	656830
Interior separate	0.1028	0.321	0.188	0.133	2.421	−1.191	1.230	657447
	in ²	in	in	in	in	in	in	
Exterior	0.000805	0.0284	0.0167	0.0118	0.2584	−0.1328	0.1256	12132817
Interior	0.000838	0.0289	0.0137	0.0110	0.1421	−0.0669	0.0752	656830
Interior separate	0.000159	0.0126	0.0074	0.0052	0.0953	−0.0469	0.0484	657447

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	2.361	1.099	4.108	0.338	0.141	0.528	0.617	0.292	0.961	3243	0.050
Interior	1.823	1.507	2.130	0.193	0.165	0.251	0.330	0.285	0.408	100	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0930	0.0433	0.1617	0.0133	0.0055	0.0208	0.0243	0.0115	0.0378	3243	0.0020
Interior	0.0718	0.0593	0.0839	0.0076	0.0065	0.0099	0.0130	0.0112	0.0161	100	0.0020



MV002



Petrie Museum, CC BY-NC-SA



Artifact Precision Report, version 01.20	
Material	Breccia
Description	Squat red and white breccia vase, with two cylindrical handles (Petrie type 5). Bought by Petrie at Abydos (Journal 1902-3, p.7)
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Bought by Petrie at Abydos
Precision Score	Exterior 1.02
Scan note	The scanned exterior surface of of the vessel contains a high number of faulty scan datapoints, this will influence and distort circularity measurements of the object.

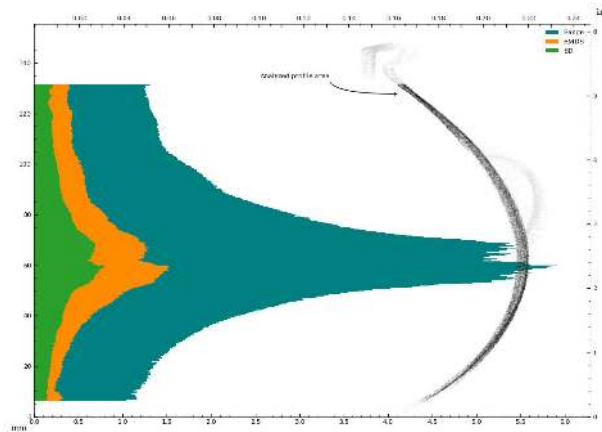
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.9798	0.990	0.575	0.406	6.052	−3.567	2.485	5265780
Interior	2.5006	1.581	0.883	0.495	14.618	−4.222	10.397	316819
	in ²	in	in	in	in	in	in	
Exterior	0.001519	0.0390	0.0226	0.0160	0.2383	−0.1404	0.0978	5265780
Interior	0.003876	0.0623	0.0348	0.0195	0.5755	−0.1662	0.4093	316819

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.859	0.583	5.988	0.269	0.117	0.802	0.533	0.197	1.536	2505	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0732	0.0229	0.2357	0.0106	0.0046	0.0316	0.0210	0.0078	0.0605	2505	0.0020



MV003



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Artifact Precision Report, version 01.20

Material

Andesite Porphyry

Description

Andesite porphyry barrel vase with two lug handles, perforated horizontally, and flat base. Rim chipped.

Collection

Petrie Museum of Egyptian Archaeology

Provenance

Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie

Precision Score

Exterior 24, Interior 748

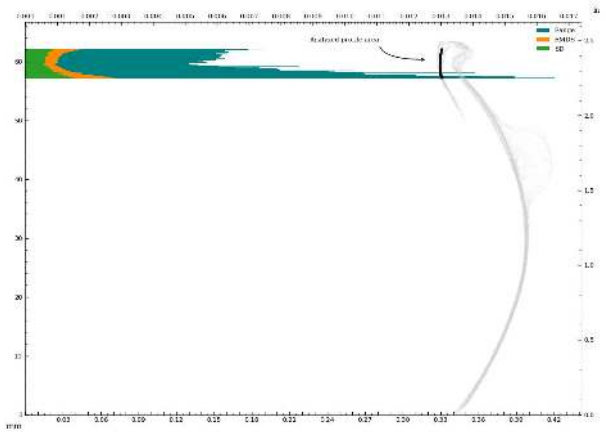
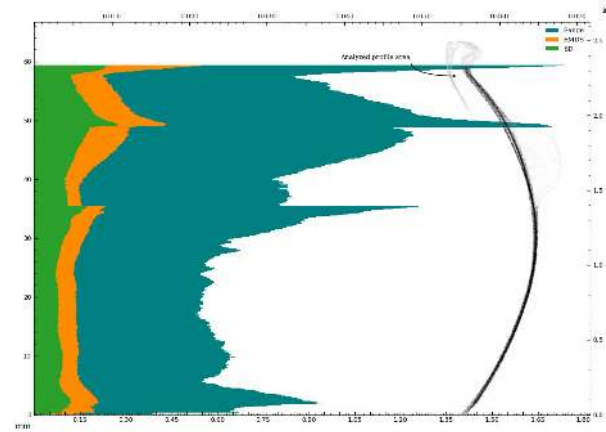
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0422	0.205	0.135	0.073	2.249	−0.806	1.443	1817152
Interior	0.1039	0.322	0.175	0.140	1.412	−0.743	0.669	92728
Interior separate	0.0013	0.037	0.024	0.014	0.497	−0.231	0.267	91395
	in ²	in	in	in	in	in	in	
Exterior	0.000065	0.0081	0.0053	0.0029	0.0885	−0.0317	0.0568	1817152
Interior	0.000161	0.0127	0.0069	0.0055	0.0556	−0.0292	0.0263	92728
Interior separate	0.000002	0.0014	0.0009	0.0005	0.0196	−0.0091	0.0105	91395

Circularity measurements

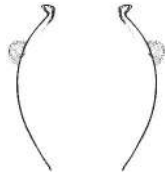
Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.778	0.376	1.735	0.102	0.054	0.281	0.160	0.127	0.549	1185	0.050
Interior	0.159	0.122	0.421	0.018	0.013	0.040	0.030	0.022	0.071	100	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0306	0.0148	0.0683	0.0040	0.0021	0.0111	0.0063	0.0050	0.0216	1185	0.0020
Interior	0.0063	0.0048	0.0166	0.0007	0.0005	0.0016	0.0012	0.0009	0.0028	100	0.0020



MV004



Petrie Museum, CC BY-NC-SA



Artifact Precision Report, version 01.20	
Material	Basalt
Description	Basalt barrel vase, with two lug handles perforated horizontally, flat base, rim chipped.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 72, Interior 109

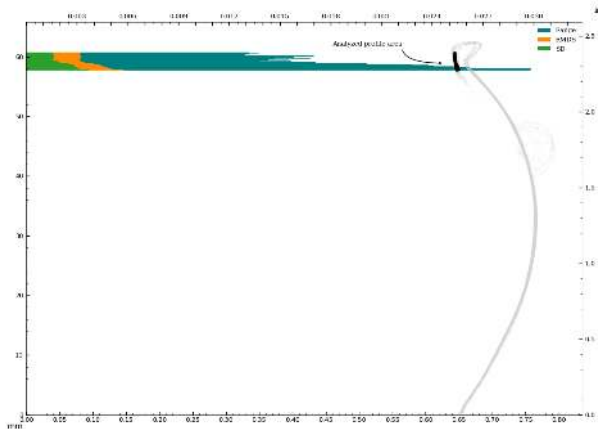
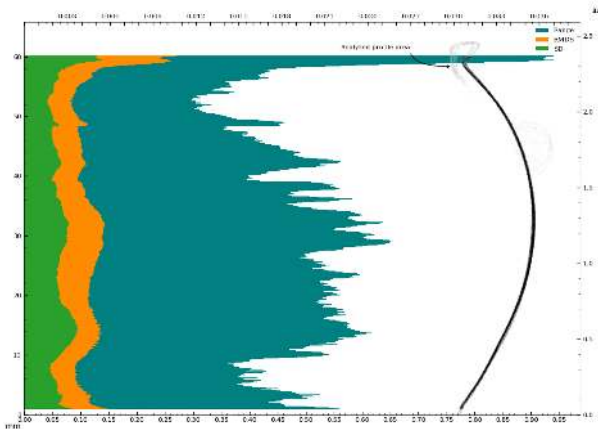
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0139	0.118	0.068	0.047	1.104	−0.496	0.609	1589197
Interior	0.1006	0.317	0.155	0.131	1.316	−0.591	0.725	46157
Interior separate	0.0092	0.096	0.057	0.035	0.805	−0.352	0.453	46174
	in ²	in	in	in	in	in	in	
Exterior	0.000022	0.0046	0.0027	0.0018	0.0435	−0.0195	0.0240	1589197
Interior	0.000156	0.0125	0.0061	0.0052	0.0518	−0.0233	0.0285	46157
Interior separate	0.000014	0.0038	0.0022	0.0014	0.0317	−0.0139	0.0178	46174

Circularity measurements

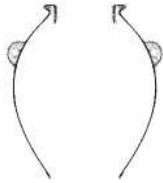
Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.502	0.290	0.940	0.061	0.040	0.138	0.114	0.080	0.269	1184	0.050
Interior	0.429	0.303	0.795	0.046	0.039	0.093	0.083	0.077	0.147	60	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0198	0.0114	0.0370	0.0024	0.0016	0.0054	0.0045	0.0032	0.0106	1184	0.0020
Interior	0.0169	0.0119	0.0313	0.0018	0.0015	0.0037	0.0033	0.0030	0.0058	60	0.0020



MV005



Petrie Museum, CC BY-NC-SA



Artifact Precision Report, version 01.20	
Material	Limestone
Description	White-veined grey limestone barrel vase; two cylindrical handles and flat base. Rim chipped.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 114, Interior 134
Scan note	Only half of interior opening captured

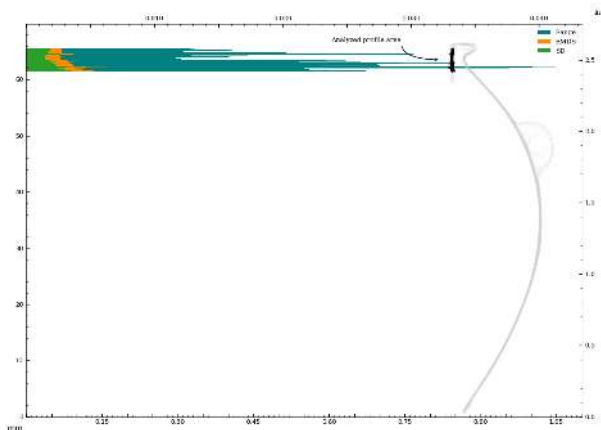
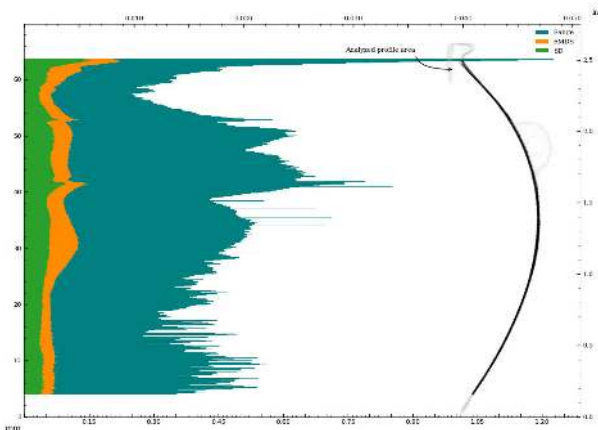
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0088	0.094	0.059	0.037	1.275	−0.743	0.532	1649358
Interior	0.0997	0.316	0.169	0.154	1.766	−1.062	0.704	46818
Interior separate	0.0075	0.087	0.058	0.033	1.171	−0.804	0.367	47945
	in ²	in	in	in	in	in	in	
Exterior	0.000014	0.0037	0.0023	0.0015	0.0502	−0.0293	0.0209	1649358
Interior	0.000154	0.0124	0.0067	0.0061	0.0695	−0.0418	0.0277	46818
Interior separate	0.000012	0.0034	0.0023	0.0013	0.0461	−0.0317	0.0145	47945

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		
Exterior	0.447	0.248	1.227	0.053	0.033	0.153	0.092	0.056	0.221	1195	0.050
Interior	0.532	0.271	1.049	0.047	0.036	0.117	0.080	0.064	0.164	80	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0176	0.0098	0.0483	0.0021	0.0013	0.0060	0.0036	0.0022	0.0087	1195	0.0020
Interior	0.0209	0.0107	0.0413	0.0019	0.0014	0.0046	0.0032	0.0025	0.0064	80	0.0020



MV006



Artifact Precision Report, version 01.20

Material Basalt

Description Basalt cylinder vase, tapering at base, rim damaged

Collection Petrie Museum of Egyptian Archaeology

Provenance Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie

Precision Score Exterior 11, Interior 274

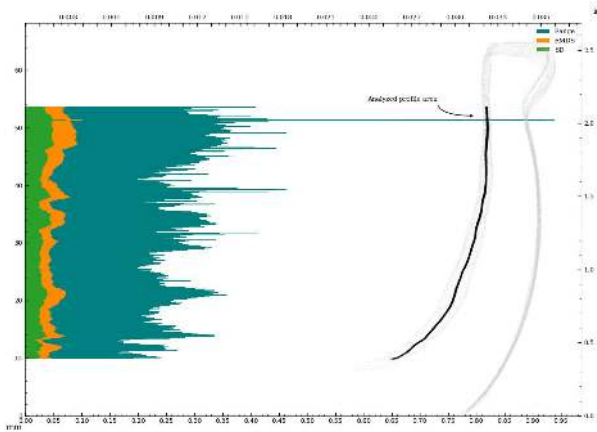
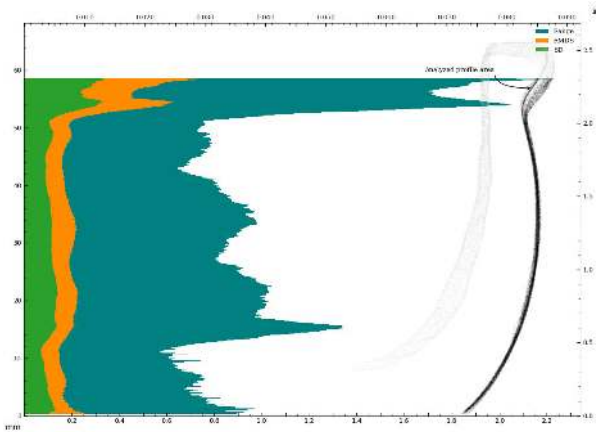
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0912	0.302	0.222	0.085	2.907	−1.592	1.315	2252063
Interior	0.7930	0.890	0.461	0.357	4.013	−2.048	1.966	1190605
Interior separate	0.0036	0.060	0.037	0.023	1.128	−0.802	0.326	1191112
	in ²	in	in	in	in	in	in	
Exterior	0.000141	0.0119	0.0087	0.0034	0.1144	−0.0627	0.0518	2252063
Interior	0.001229	0.0351	0.0181	0.0141	0.1580	−0.0806	0.0774	1190605
Interior separate	0.000006	0.0024	0.0015	0.0009	0.0444	−0.0316	0.0128	1191112

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.854	0.567	2.229	0.114	0.068	0.335	0.193	0.143	0.728	1167	0.050
Interior	0.270	0.150	0.939	0.031	0.018	0.073	0.055	0.028	0.101	882	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0336	0.0223	0.0878	0.0045	0.0027	0.0132	0.0076	0.0056	0.0287	1167	0.0020
Interior	0.0106	0.0059	0.0370	0.0012	0.0007	0.0029	0.0021	0.0011	0.0040	882	0.0020



MV007



Artifact Precision Report, version 01.20

Material Granite

Description Black and white granite squat brim vase. Part upper missing

Collection Petrie Museum of Egyptian Archaeology

Provenance Petrie Museum of Egyptian Archaeology

Precision Score Exterior 19

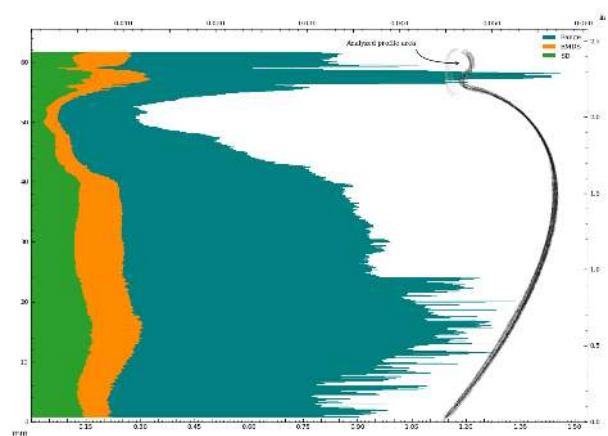
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0531	0.231	0.137	0.105	1.750	−0.883	0.868	1953524
Interior	0.1315	0.363	0.233	0.158	2.175	−0.778	1.396	98016
	in ²	in	in	in	in	in	in	
Exterior	0.000082	0.0091	0.0054	0.0041	0.0689	−0.0348	0.0342	1953524
Interior	0.000204	0.0143	0.0092	0.0062	0.0856	−0.0306	0.0550	98016

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.908	0.276	1.462	0.128	0.033	0.172	0.248	0.071	0.331	1221	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0357	0.0109	0.0575	0.0051	0.0013	0.0068	0.0098	0.0028	0.0130	1221	0.0020



MV008



Petrie Museum, CC BY-NC-SA



Artifact Precision Report, version 01.20	
Material	Diorite
Description	Black spotted white diorite vase. Rim area broken away
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 200
Scan note	Interior surface of the scan not analyzed due to damage on the artifact.

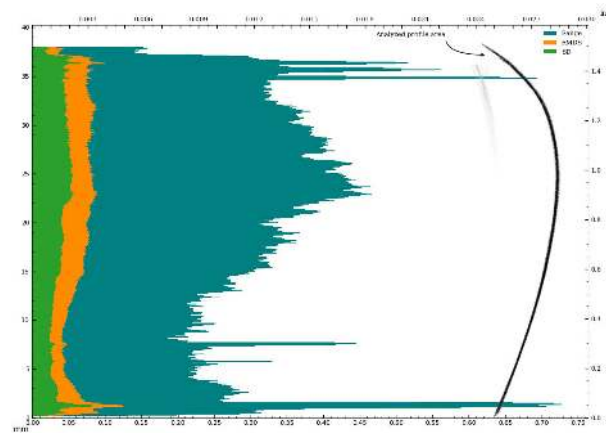
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0050	0.071	0.043	0.028	1.084	−0.461	0.623	684997
Interior	0.0656	0.256	0.174	0.097	2.005	−0.761	1.244	61789
	in ²	in	in	in	in	in	in	
Exterior	0.000008	0.0028	0.0017	0.0011	0.0427	−0.0182	0.0245	684997
Interior	0.000102	0.0101	0.0069	0.0038	0.0789	−0.0300	0.0490	61789

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.326	0.131	0.728	0.041	0.017	0.083	0.074	0.026	0.126	756	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0128	0.0052	0.0286	0.0016	0.0007	0.0033	0.0029	0.0010	0.0049	756	0.0020



MV009



Petrie Museum, CC BY-NC-SA



Artifact Precision Report, version 01.20

Material Diorite

Description Banded grey Diorite gneiss bowl, part of rim missing.

Collection Petrie Museum of Egyptian Archaeology

Provenance Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie

Precision Score Exterior 15, Interior 63

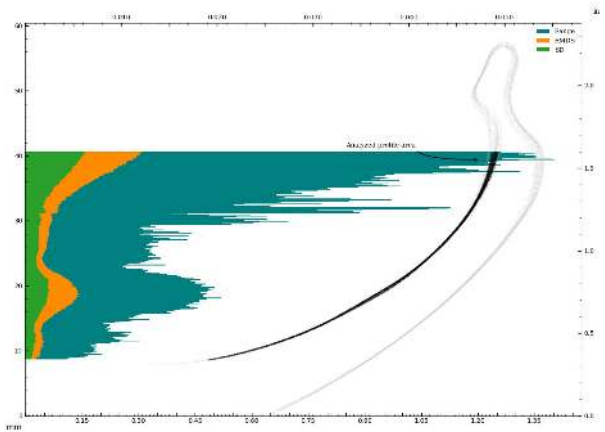
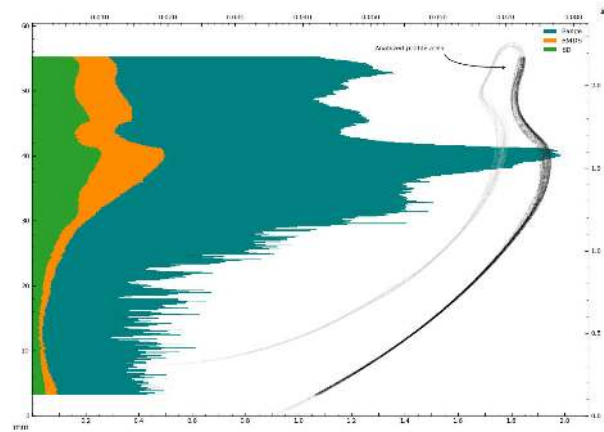
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0668	0.258	0.193	0.071	2.036	−1.218	0.818	3980685
Interior	0.0474	0.218	0.168	0.052	2.073	−1.023	1.049	2504454
Interior separate	0.0158	0.126	0.089	0.039	1.473	−0.723	0.751	2484789
	in ²	in	in	in	in	in	in	
Exterior	0.000104	0.0102	0.0076	0.0028	0.0801	−0.0479	0.0322	3980685
Interior	0.000073	0.0086	0.0066	0.0020	0.0816	−0.0403	0.0413	2504454
Interior separate	0.000025	0.0050	0.0035	0.0016	0.0580	−0.0285	0.0295	2484789

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.055	0.241	1.989	0.131	0.021	0.259	0.184	0.034	0.498	1042	0.050
Interior	0.424	0.107	1.403	0.040	0.014	0.158	0.077	0.027	0.310	637	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0415	0.0095	0.0783	0.0052	0.0008	0.0102	0.0072	0.0014	0.0196	1042	0.0020
Interior	0.0167	0.0042	0.0553	0.0016	0.0006	0.0062	0.0030	0.0011	0.0122	637	0.0020



MV010



Petrie Museum, CC BY-NC-SA



Artifact Precision Report, version 01.20	
Material	Calcite (Egyptian Alabaster)
Description	Calcite (Egyptian alabaster) keeled vase, from the lower chamber in the tomb of the governor May at Qau (tomb no. 1456).Calcite (Egyptian alabaster) keeled vase, from the lower chamber in the tomb of the governor May at Qau (tomb no. 1456).
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 1.17, Interior 11

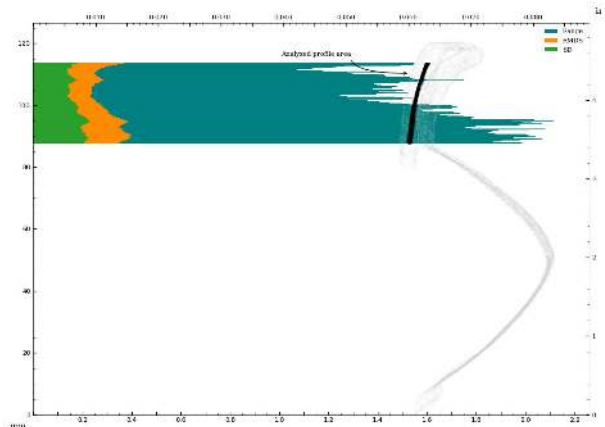
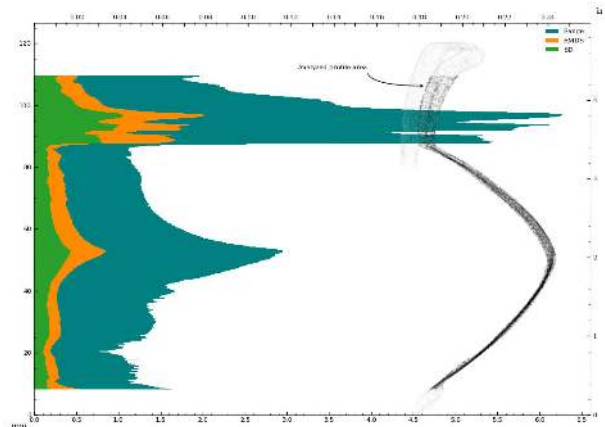
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.8522	0.923	0.688	0.256	7.618	−4.091	3.527	10583391
Interior	6.0097	2.451	1.084	0.872	8.785	−3.865	4.920	1514659
Interior separate	0.0948	0.308	0.185	0.115	2.418	−1.187	1.231	1506354
	in ²	in	in	in	in	in	in	
Exterior	0.001321	0.0363	0.0271	0.0101	0.2999	−0.1611	0.1388	10583391
Interior	0.009315	0.0965	0.0427	0.0344	0.3459	−0.1522	0.1937	1514659
Interior separate	0.000147	0.0121	0.0073	0.0045	0.0952	−0.0467	0.0485	1506354

Circularity measurements

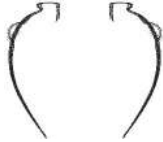
Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.451	0.749	6.286	0.188	0.102	1.057	0.370	0.182	2.030	2028	0.050
Interior	1.568	1.052	2.151	0.172	0.134	0.232	0.285	0.227	0.398	520	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0571	0.0295	0.2475	0.0074	0.0040	0.0416	0.0145	0.0072	0.0799	2028	0.0020
Interior	0.0617	0.0414	0.0847	0.0068	0.0053	0.0091	0.0112	0.0089	0.0157	520	0.0020



MV011



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Artifact Precision Report, version 01.20

Material	Porphyry
Description	Black and white porphyry vase; flat slightly rimmed base, two perforated longitudinal handles on rounded shoulder, constricted to flat-topped rim. From the royal tomb of King Den in the Umm el-Qa'ab at Abydos. Possibly an heirloom in a later context.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 1.81, Interior 40
Scan note	Scan contains invalid datapoints all along the exterior side, this scan error has been corrected, possibly also removing some valid datapoints in the process.

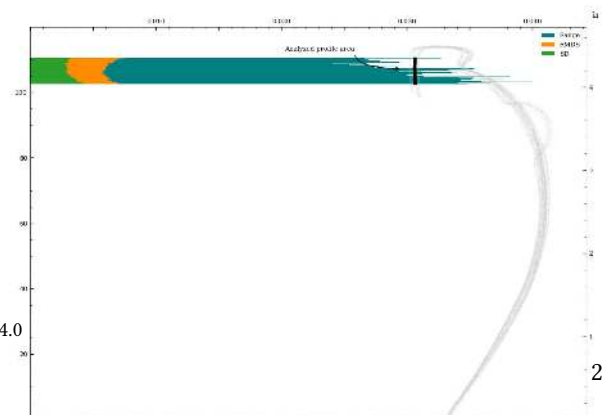
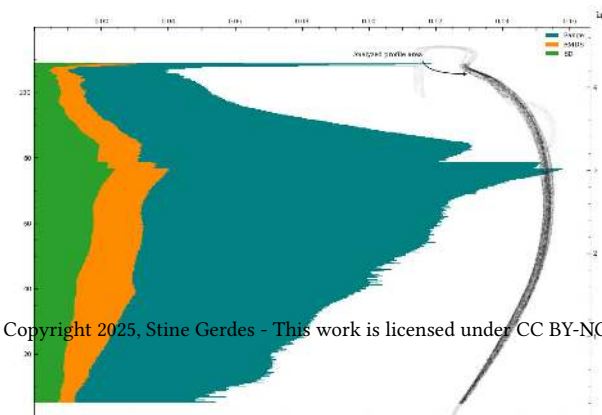
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.5522	0.743	0.419	0.300	4.092	-2.286	1.806	7372669
Interior	0.8970	0.947	0.447	0.372	3.258	-1.771	1.487	236003
Interior separate	0.0252	0.159	0.083	0.056	1.186	-0.587	0.600	236001
	in ²	in	in	in	in	in	in	
Exterior	0.000856	0.0293	0.0165	0.0118	0.1611	-0.0900	0.0711	7372669
Interior	0.001390	0.0373	0.0176	0.0147	0.1283	-0.0697	0.0586	236003
Interior separate	0.000039	0.0062	0.0033	0.0022	0.0467	-0.0231	0.0236	236001

Circularity measurements

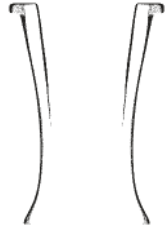
Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	2.404	0.986	4.020	0.316	0.113	0.616	0.702	0.247	1.023	2077	0.050
Interior	0.786	0.609	1.071	0.076	0.067	0.109	0.156	0.143	0.191	160	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0946	0.0388	0.1583	0.0124	0.0045	0.0242	0.0276	0.0097	0.0403	2077	0.0020
Interior	0.0309	0.0240	0.0422	0.0030	0.0026	0.0043	0.0061	0.0056	0.0075	160	0.0020



MV012



Petrie Museum, CC BY-NC-SA



Artifact Precision Report, version 01.20	
Material	Basalt
Description	Basalt cylinder jar with flared base, one large chip on rim, several smaller on base. Otherwise intact.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology
Precision Score	Exterior 6, Interior 33

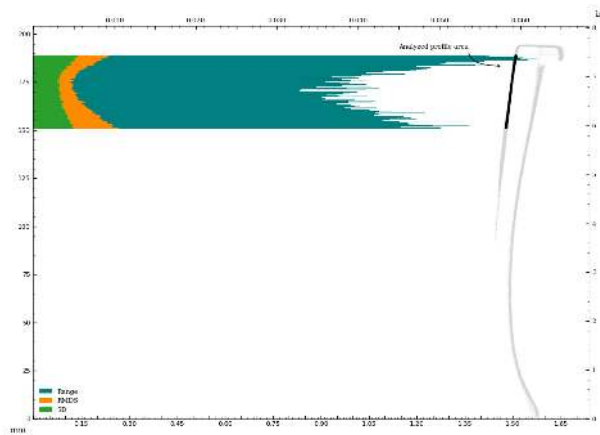
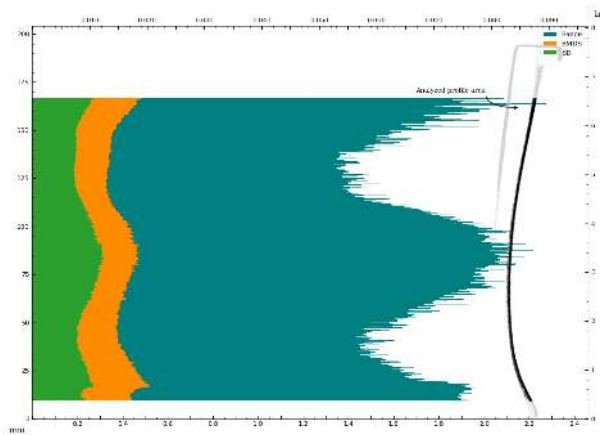
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.1682	0.410	0.245	0.154	2.810	−1.737	1.073	7010564
Interior	0.1453	0.381	0.196	0.150	2.306	−1.149	1.156	2656349
Interior separate	0.0299	0.173	0.107	0.068	1.707	−0.823	0.884	2657943
	in ²	in	in	in	in	in	in	
Exterior	0.000261	0.0161	0.0096	0.0061	0.1106	−0.0684	0.0422	7010564
Interior	0.000225	0.0150	0.0077	0.0059	0.0908	−0.0453	0.0455	2656349
Interior separate	0.000046	0.0068	0.0042	0.0027	0.0672	−0.0324	0.0348	2657943

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.700	1.306	2.344	0.226	0.178	0.317	0.397	0.316	0.523	3151	0.050
Interior	1.037	0.799	1.656	0.093	0.071	0.145	0.158	0.109	0.268	760	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0669	0.0514	0.0923	0.0089	0.0070	0.0125	0.0156	0.0124	0.0206	3151	0.0020
Interior	0.0408	0.0314	0.0652	0.0036	0.0028	0.0057	0.0062	0.0043	0.0105	760	0.0020



MV013



Petrie Museum, CC BY-NC-SA



Artifact Precision Report, version 01.20

Material Diorite

Description Slender diorite cylinder jar with sharply flared base and protruding rim. Large chip on base and approximately one third of rim missing.

Collection Petrie Museum of Egyptian Archaeology

Provenance Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie

Precision Score Exterior 4, Interior 6

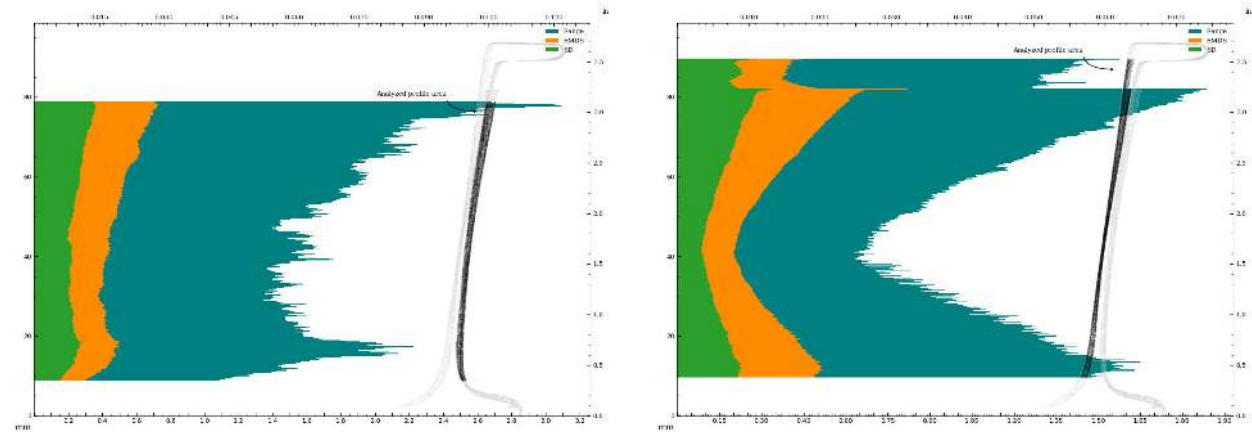
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.2603	0.510	0.270	0.211	3.151	−1.775	1.375	1784377
Interior	0.2717	0.521	0.287	0.199	2.768	−1.272	1.496	1656151
Interior separate	0.1590	0.399	0.212	0.151	2.096	−0.967	1.129	1657531
	in ²	in	in	in	in	in	in	
Exterior	0.000403	0.0201	0.0106	0.0083	0.1240	−0.0699	0.0541	1784377
Interior	0.000421	0.0205	0.0113	0.0078	0.1090	−0.0501	0.0589	1656151
Interior separate	0.000246	0.0157	0.0084	0.0059	0.0825	−0.0381	0.0444	1657531

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.628	1.056	3.107	0.232	0.145	0.360	0.458	0.291	0.729	1400	0.050
Interior	1.183	0.616	1.886	0.165	0.084	0.338	0.372	0.196	0.819	1600	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0641	0.0416	0.1223	0.0091	0.0057	0.0142	0.0180	0.0115	0.0287	1400	0.0020
Interior	0.0466	0.0243	0.0743	0.0065	0.0033	0.0133	0.0146	0.0077	0.0322	1600	0.0020



MV014



Petrie Museum, CC BY-NC-SA



Artifact Precision Report, version 01.20	
Material	Basalt
Description	Black basalt bowl. Distinct drill marks on interior base. Rim chipped but otherwise intact.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 54, Interior 243
Scan note	The scanned exterior surface of of the vessel contains a high number of faulty scan datapoints, these have been manually removed.

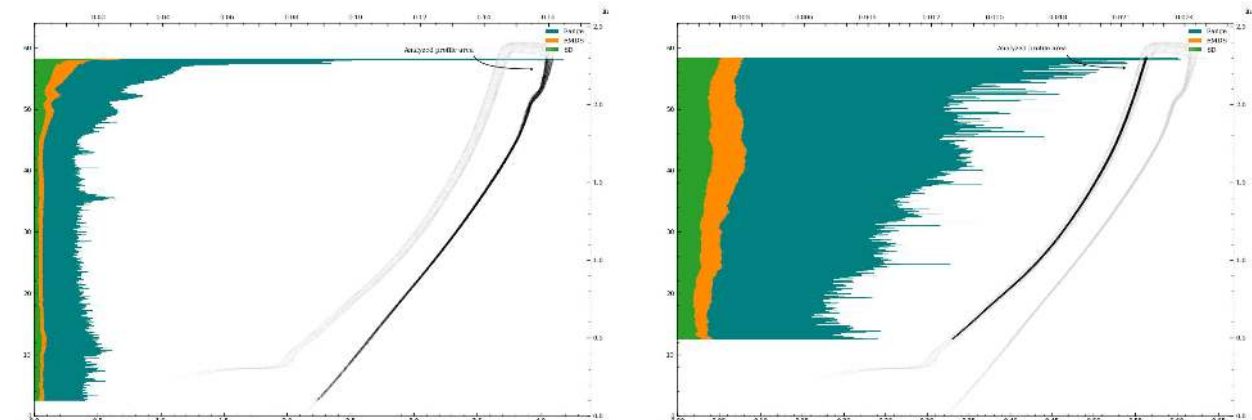
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0184	0.136	0.098	0.035	3.836	−2.908	0.928	3155164
Interior	0.2340	0.484	0.223	0.175	1.940	−0.917	1.024	2825923
Interior separate	0.0041	0.064	0.037	0.026	0.646	−0.288	0.358	2825431
	in ²	in	in	in	in	in	in	
Exterior	0.000029	0.0053	0.0039	0.0014	0.1510	−0.1145	0.0365	3155164
Interior	0.000363	0.0190	0.0088	0.0069	0.0764	−0.0361	0.0403	2825923
Interior separate	0.000006	0.0025	0.0015	0.0010	0.0254	−0.0114	0.0141	2825431

Circularity measurements

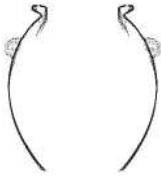
Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.397	0.290	4.186	0.041	0.032	0.511	0.073	0.060	0.676	1119	0.050
Interior	0.282	0.152	0.636	0.033	0.018	0.053	0.063	0.034	0.084	920	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0156	0.0114	0.1648	0.0016	0.0012	0.0201	0.0029	0.0024	0.0266	1119	0.0020
Interior	0.0111	0.0060	0.0250	0.0013	0.0007	0.0021	0.0025	0.0014	0.0033	920	0.0020



MV015



Petrie Museum, CC BY-NC-SA



Artifact Precision Report, version 01.20

Material Black & White Syenite, perhaps Hornblende Diorite

Description Stone vase, identified in museum register as black and white syenite, perhaps hornblende diorite (?), barrel-shaped type 39. From Naqada Tomb T 16

Collection Petrie Museum of Egyptian Archaeology

Provenance Petrie Museum of Egyptian Archaeology

Precision Score Exterior 26, Interior 9

Scan note Many scan artifacts present around the rim/neck area of the scan. These faulty verties have been removed manually.

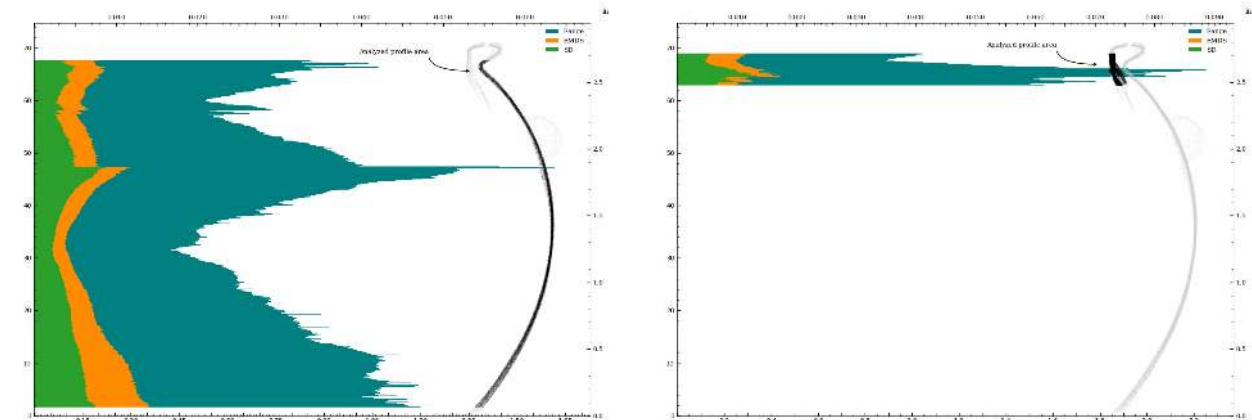
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0382	0.195	0.125	0.076	1.873	−1.265	0.607	2204479
Interior	0.2716	0.521	0.334	0.154	3.168	−0.976	2.192	112746
Interior separate	0.1072	0.327	0.223	0.105	2.731	−1.076	1.654	112386
	in ²	in	in	in	in	in	in	
Exterior	0.000059	0.0077	0.0049	0.0030	0.0737	−0.0498	0.0239	2204479
Interior	0.000421	0.0205	0.0132	0.0061	0.1247	−0.0384	0.0863	112746
Interior separate	0.000166	0.0129	0.0088	0.0042	0.1075	−0.0424	0.0651	112386

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.776	0.423	1.645	0.105	0.055	0.220	0.178	0.093	0.363	1321	0.050
Interior	1.518	0.884	2.254	0.176	0.115	0.342	0.286	0.230	0.438	120	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0305	0.0166	0.0648	0.0041	0.0022	0.0087	0.0070	0.0037	0.0143	1321	0.0020
Interior	0.0598	0.0348	0.0888	0.0069	0.0045	0.0135	0.0113	0.0091	0.0172	120	0.0020



MV016



Petrie Museum, CC BY-NC-SA



Artifact Precision Report, version 01.20	
Material	Basalt
Description	Stone vase, basalt, barrel shaped, type 55. From Naqada Tomb T 16
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 2.22, Interior 28

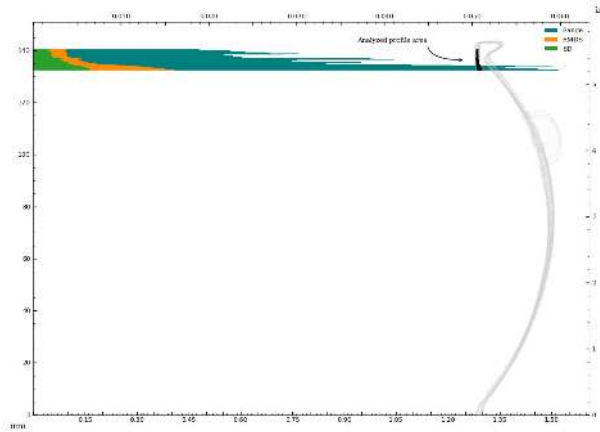
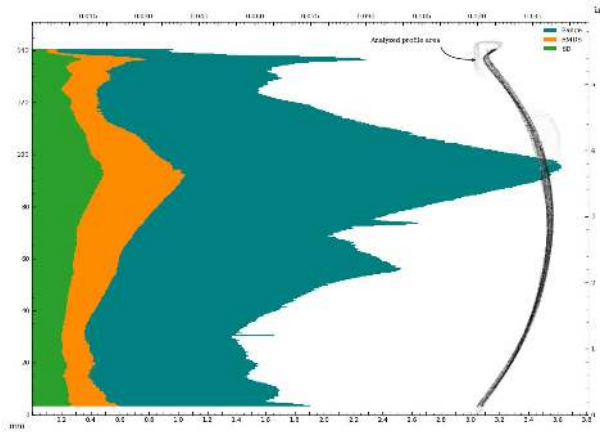
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

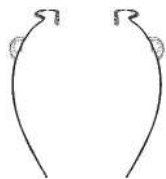
Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.4505	0.671	0.380	0.271	3.666	−2.262	1.404	7408951
Interior	0.4197	0.648	0.317	0.243	2.800	−1.203	1.598	242313
Interior separate	0.0363	0.190	0.126	0.059	1.700	−0.939	0.762	240821
	in ²	in	in	in	in	in	in	
Exterior	0.000698	0.0264	0.0150	0.0107	0.1443	−0.0890	0.0553	7408951
Interior	0.000651	0.0255	0.0125	0.0096	0.1103	−0.0474	0.0629	242313
Interior separate	0.000056	0.0075	0.0050	0.0023	0.0669	−0.0370	0.0300	240821

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	2.003	0.850	3.628	0.268	0.091	0.498	0.532	0.159	1.059	2747	0.050
Interior	0.806	0.459	1.532	0.078	0.044	0.170	0.131	0.088	0.412	160	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0788	0.0335	0.1428	0.0106	0.0036	0.0196	0.0209	0.0063	0.0417	2747	0.0020
Interior	0.0317	0.0181	0.0603	0.0031	0.0017	0.0067	0.0052	0.0035	0.0162	160	0.0020



MV017



Artifact Precision Report, version 01.20

Material Limestone

Description Vase of hard limestone, type 43, barrel shape, rim damaged. From Naqada Tomb 743.

Collection Petrie Museum of Egyptian Archaeology

Provenance Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie

Precision Score Exterior 30, Interior 342

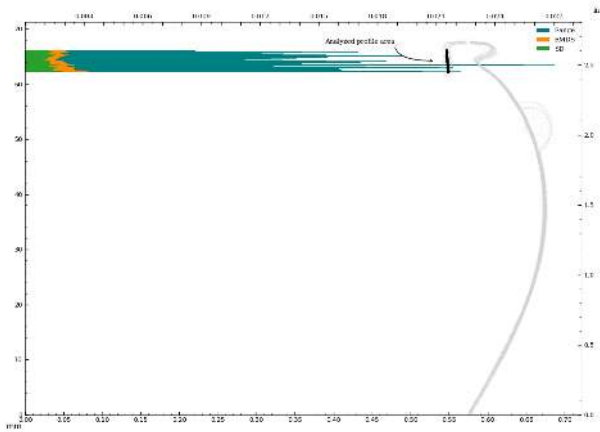
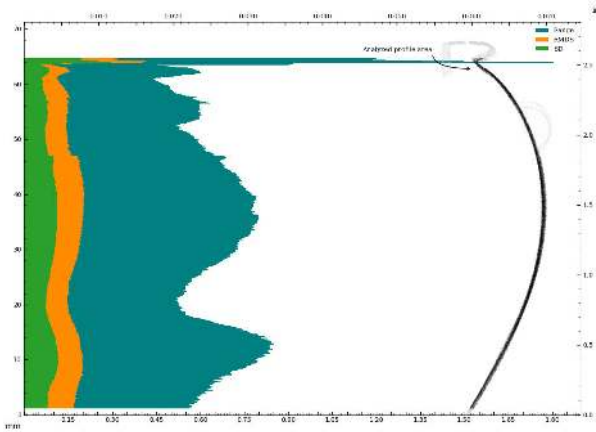
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0337	0.184	0.110	0.068	3.180	−0.930	2.250	1973350
Interior	0.2666	0.516	0.237	0.182	1.869	−0.860	1.009	56343
Interior separate	0.0029	0.054	0.035	0.019	0.833	−0.458	0.375	56114
	in ²	in	in	in	in	in	in	
Exterior	0.000052	0.0072	0.0043	0.0027	0.1252	−0.0366	0.0886	1973350
Interior	0.000413	0.0203	0.0093	0.0072	0.0736	−0.0339	0.0397	56343
Interior separate	0.000005	0.0021	0.0014	0.0008	0.0328	−0.0180	0.0148	56114

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.647	0.413	1.804	0.091	0.057	0.296	0.176	0.092	0.415	1272	0.050
Interior	0.360	0.217	0.687	0.032	0.022	0.049	0.052	0.036	0.082	80	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0255	0.0163	0.0710	0.0036	0.0023	0.0117	0.0069	0.0036	0.0163	1272	0.0020
Interior	0.0142	0.0085	0.0271	0.0013	0.0009	0.0019	0.0021	0.0014	0.0032	80	0.0020



MV018



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Artifact Precision Report, version 01.20	
Material	Diorite
Description	Vase of diorite, squat, type 15, rim and one handle broken. From Naqada Cemetery T Tomb 20
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 13, Interior 86

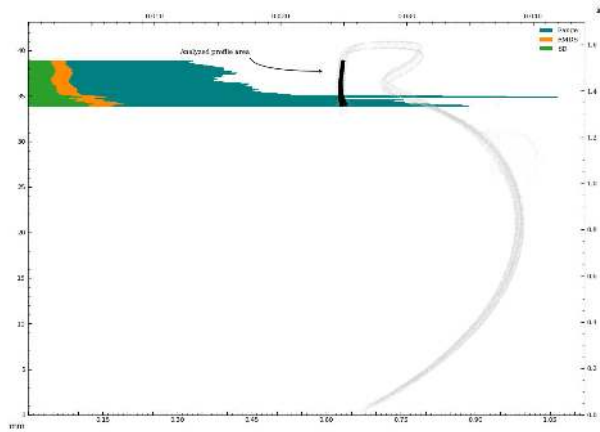
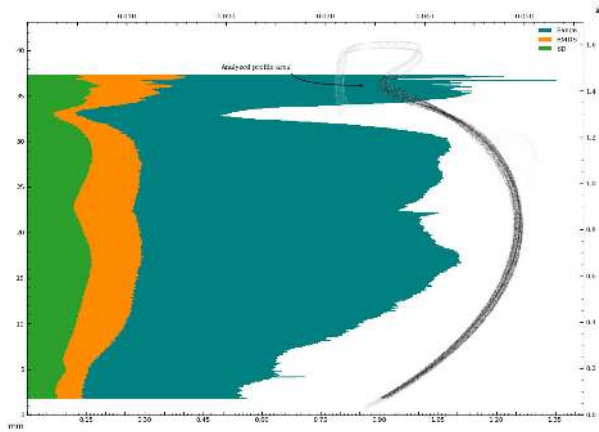
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0744	0.273	0.170	0.113	5.425	−2.752	2.673	1646185
Interior	0.1250	0.354	0.160	0.117	1.641	−0.670	0.972	97568
Interior separate	0.0117	0.108	0.071	0.035	1.443	−0.794	0.649	97171
	in ²	in	in	in	in	in	in	
Exterior	0.000115	0.0107	0.0067	0.0044	0.2136	−0.1084	0.1052	1646185
Interior	0.000194	0.0139	0.0063	0.0046	0.0646	−0.0264	0.0383	97568
Interior separate	0.000018	0.0043	0.0028	0.0014	0.0568	−0.0312	0.0256	97171

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.006	0.491	1.354	0.137	0.065	0.193	0.273	0.118	0.409	711	0.050
Interior	0.430	0.323	1.066	0.053	0.044	0.131	0.088	0.072	0.193	100	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0396	0.0193	0.0533	0.0054	0.0026	0.0076	0.0107	0.0047	0.0161	711	0.0020
Interior	0.0169	0.0127	0.0420	0.0021	0.0017	0.0052	0.0035	0.0028	0.0076	100	0.0020



MV019



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Artifact Precision Report, version 01.20

Material

Porphyry

Description

Stone vase, type 99, porphyry, rim damaged. From Naqada Cemetery B Tomb 99

Collection

Petrie Museum of Egyptian Archaeology

Provenance

Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie

Precision Score

Exterior 14, Interior 21

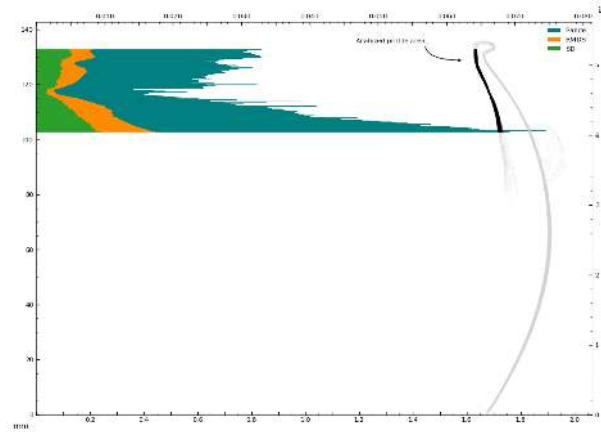
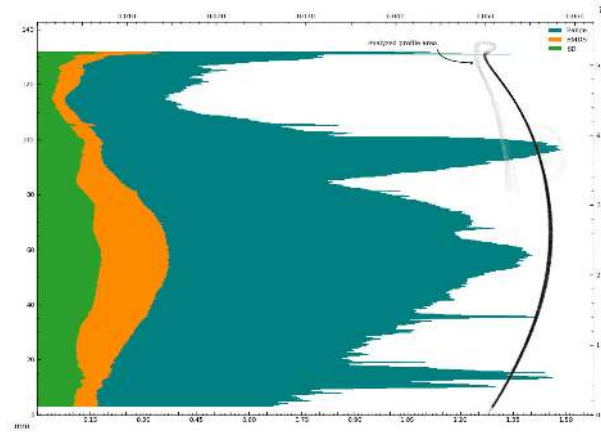
Selected measurements shown below, see full artifact report for further details.

Surface Variability

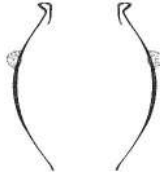
Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0701	0.265	0.158	0.113	2.625	−1.244	1.381	6855158
Interior	0.1894	0.435	0.238	0.162	2.253	−0.779	1.473	783536
Interior separate	0.0485	0.220	0.139	0.085	1.927	−0.686	1.241	783508
	in ²	in	in	in	in	in	in	
Exterior	0.000109	0.0104	0.0062	0.0044	0.1033	−0.0490	0.0544	6855158
Interior	0.000294	0.0171	0.0094	0.0064	0.0887	−0.0307	0.0580	783536
Interior separate	0.000075	0.0087	0.0055	0.0033	0.0759	−0.0270	0.0488	783508

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.073	0.355	1.502	0.123	0.040	0.189	0.229	0.078	0.377	2581	0.050
Interior	0.737	0.345	1.968	0.096	0.035	0.237	0.190	0.063	0.443	600	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0422	0.0140	0.0591	0.0048	0.0016	0.0075	0.0090	0.0031	0.0148	2581	0.0020
Interior	0.0290	0.0136	0.0775	0.0038	0.0014	0.0093	0.0075	0.0025	0.0175	600	0.0020



MV020



Artifact Precision Report, version 01.20	
Material	Basalt
Description	Stone vase, basalt, type B 37. from Naqada Tomb 100.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 9, Interior 464

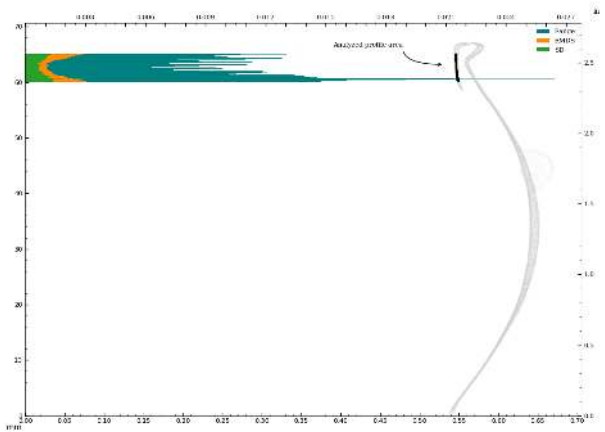
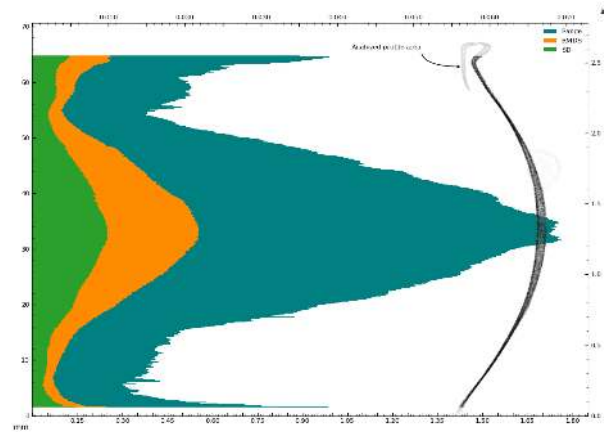
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.1116	0.334	0.213	0.128	1.819	−0.851	0.968	1801750
Interior	0.0415	0.204	0.093	0.075	0.889	−0.455	0.433	85440
Interior separate	0.0022	0.046	0.029	0.018	0.721	−0.456	0.265	85451
	in ²	in	in	in	in	in	in	
Exterior	0.000173	0.0132	0.0084	0.0050	0.0716	−0.0335	0.0381	1801750
Interior	0.000064	0.0080	0.0037	0.0030	0.0350	−0.0179	0.0171	85440
Interior separate	0.000003	0.0018	0.0011	0.0007	0.0284	−0.0180	0.0104	85451

Circularity measurements

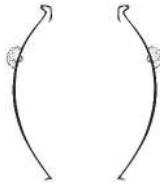
Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.844	0.287	1.764	0.108	0.034	0.255	0.237	0.067	0.556	1265	0.050
Interior	0.249	0.159	0.672	0.022	0.015	0.046	0.037	0.026	0.078	100	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0332	0.0113	0.0694	0.0043	0.0013	0.0100	0.0093	0.0026	0.0219	1265	0.0020
Interior	0.0098	0.0062	0.0264	0.0009	0.0006	0.0018	0.0015	0.0010	0.0031	100	0.0020



MV021



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Artifact Precision Report, version 01.20	
Material	Grey Basalt
Description	Grey basalt vase with two perforated handles. From Naqada Tomb 1207
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 10, Interior 91

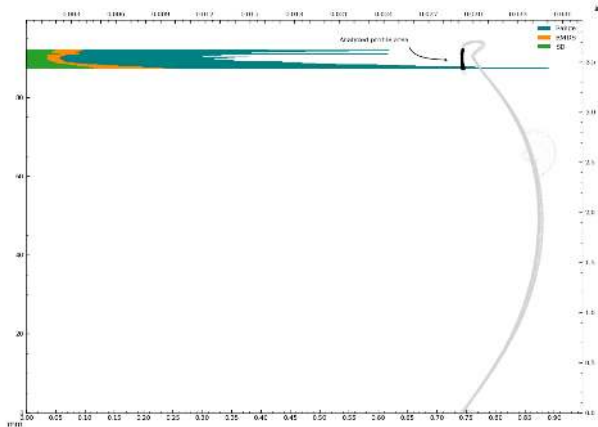
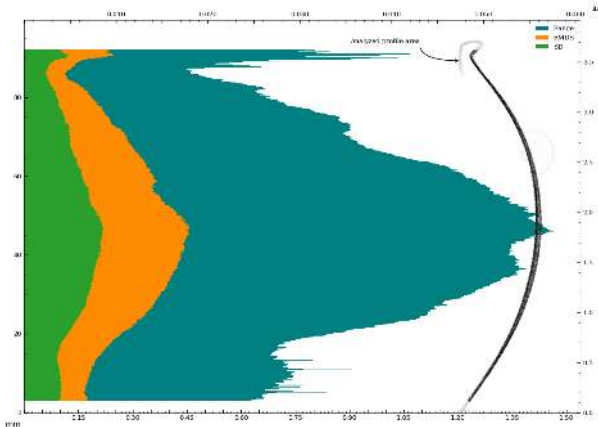
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.1040	0.322	0.177	0.128	1.490	−0.671	0.819	3475413
Interior	0.0270	0.164	0.088	0.064	1.110	−0.408	0.702	101142
Interior separate	0.0110	0.105	0.070	0.036	1.166	−0.580	0.586	100930
	in ²	in	in	in	in	in	in	
Exterior	0.000161	0.0127	0.0070	0.0050	0.0587	−0.0264	0.0322	3475413
Interior	0.000042	0.0065	0.0035	0.0025	0.0437	−0.0161	0.0276	101142
Interior separate	0.000017	0.0041	0.0028	0.0014	0.0459	−0.0228	0.0231	100930

Circularity measurements

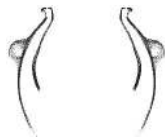
Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.982	0.441	1.465	0.132	0.056	0.218	0.301	0.109	0.459	1781	0.050
Interior	0.447	0.273	0.903	0.044	0.034	0.122	0.085	0.056	0.251	100	0.050
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0387	0.0174	0.0577	0.0052	0.0022	0.0086	0.0119	0.0043	0.0181	1781	0.0020
Interior	0.0176	0.0107	0.0356	0.0017	0.0013	0.0048	0.0033	0.0022	0.0099	100	0.0020



MV003b



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Artifact Precision Report, version 01.20	
Material	Andesite Porphyry
Description	Andesite porphyry barrel vase with two lug handles, perforated horizontally, and flat base. Rim chipped.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 19, Interior 124

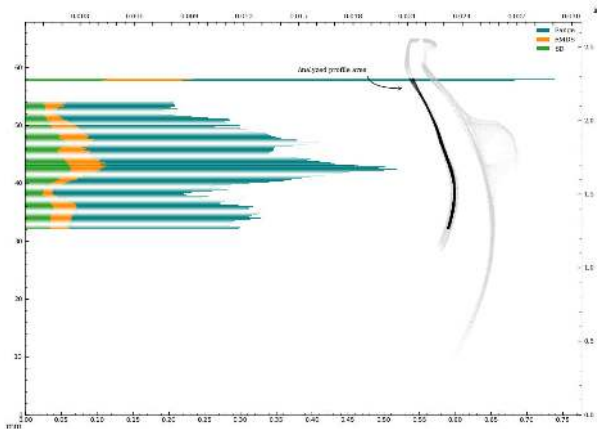
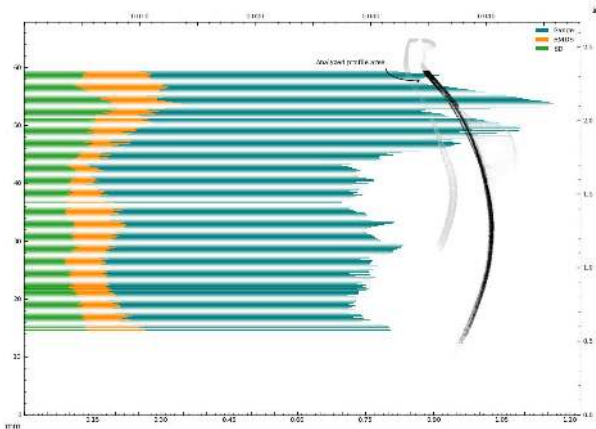
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0522	0.228	0.125	0.098	1.243	−0.613	0.629	79182
Interior	0.1409	0.375	0.170	0.135	1.346	−0.657	0.689	41298
Interior separate	0.0080	0.090	0.063	0.031	0.669	−0.257	0.412	9498
	in ²	in	in	in	in	in	in	
Exterior	0.000081	0.0090	0.0049	0.0039	0.0489	−0.0242	0.0248	79182
Interior	0.000218	0.0148	0.0067	0.0053	0.0530	−0.0259	0.0271	41298
Interior separate	0.000012	0.0035	0.0025	0.0012	0.0263	−0.0101	0.0162	9498

Circularity measurements

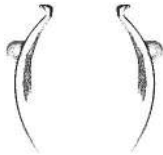
Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.768	0.651	1.164	0.111	0.084	0.205	0.199	0.133	0.377	223	0.200
Interior	0.313	0.189	0.739	0.037	0.023	0.113	0.069	0.038	0.235	112	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0302	0.0256	0.0458	0.0044	0.0033	0.0081	0.0078	0.0052	0.0149	223	0.0079
Interior	0.0123	0.0074	0.0291	0.0014	0.0009	0.0045	0.0027	0.0015	0.0092	112	0.0079



MV004b



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Artifact Precision Report, version 01.20	
Material	Basalt
Description	Basalt barrel vase, with two lug handles perforated horizontally, flat base, rim chipped.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 70, Interior 83

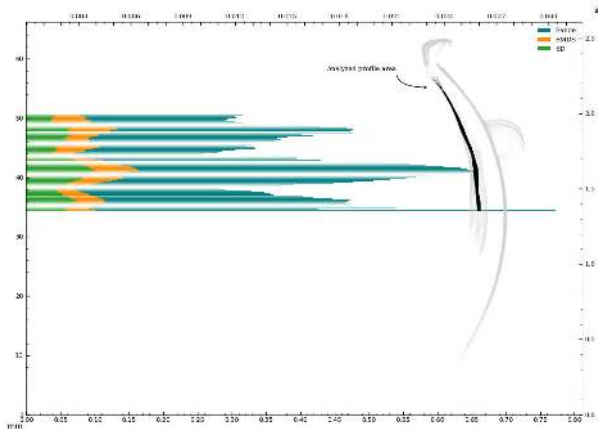
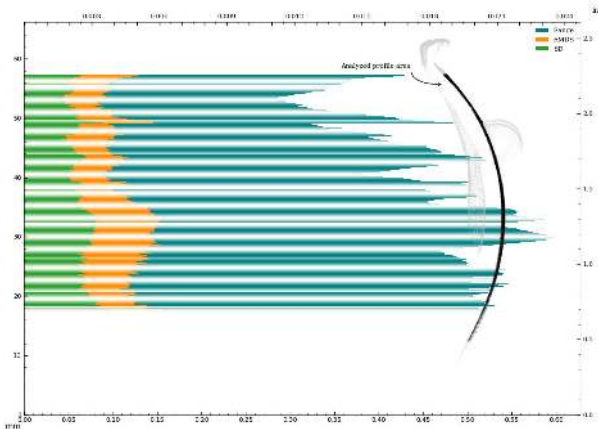
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0143	0.120	0.067	0.049	0.765	−0.379	0.386	60415
Interior	0.5374	0.733	0.384	0.292	3.019	−1.494	1.524	23593
Interior separate	0.0120	0.110	0.068	0.044	1.040	−0.486	0.554	23032
	in ²	in	in	in	in	in	in	
Exterior	0.000022	0.0047	0.0026	0.0019	0.0301	−0.0149	0.0152	60415
Interior	0.000833	0.0289	0.0151	0.0115	0.1188	−0.0588	0.0600	23593
Interior separate	0.000019	0.0043	0.0027	0.0017	0.0409	−0.0191	0.0218	23032

Circularity measurements

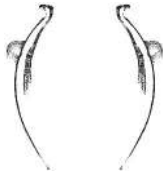
Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.484	0.285	0.597	0.063	0.041	0.094	0.116	0.080	0.158	198	0.200
Interior	0.403	0.258	0.773	0.059	0.035	0.100	0.102	0.069	0.164	82	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0191	0.0112	0.0235	0.0025	0.0016	0.0037	0.0046	0.0031	0.0062	198	0.0079
Interior	0.0159	0.0102	0.0304	0.0023	0.0014	0.0039	0.0040	0.0027	0.0065	82	0.0079



MV004c



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Artifact Precision Report, version 01.20	
Material	Basalt
Description	Basalt barrel vase, with two lug handles perforated horizontally, flat base, rim chipped.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 70, Interior 79

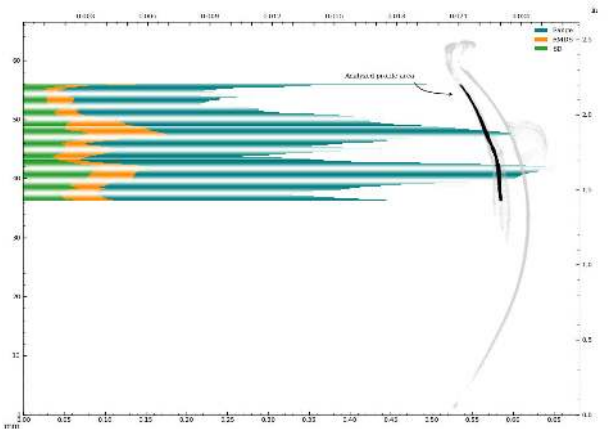
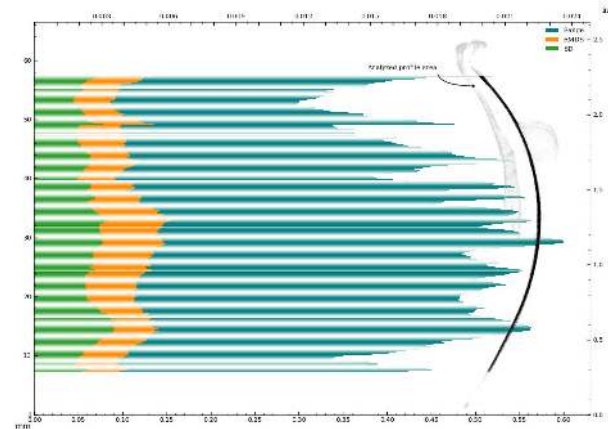
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0143	0.119	0.066	0.048	0.737	−0.385	0.352	180630
Interior	0.3703	0.609	0.329	0.240	2.878	−1.293	1.586	79801
Interior separate	0.0127	0.113	0.066	0.043	0.723	−0.372	0.351	78369
	in ²	in	in	in	in	in	in	
Exterior	0.000022	0.0047	0.0026	0.0019	0.0290	−0.0152	0.0139	180630
Interior	0.000574	0.0240	0.0130	0.0094	0.1133	−0.0509	0.0624	79801
Interior separate	0.000020	0.0044	0.0026	0.0017	0.0284	−0.0146	0.0138	78369

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.487	0.296	0.600	0.061	0.044	0.092	0.116	0.085	0.156	250	0.200
Interior	0.413	0.217	0.649	0.052	0.025	0.103	0.100	0.042	0.176	101	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0192	0.0116	0.0236	0.0024	0.0017	0.0036	0.0046	0.0034	0.0061	250	0.0079
Interior	0.0163	0.0086	0.0255	0.0020	0.0010	0.0041	0.0040	0.0016	0.0069	101	0.0079



MV015b



Petrie Museum, CC BY-NC-SA



Artifact Precision Report, version 01.20	
Material	Black & White Syenite, perhaps Hornblende Diorite
Description	Stone vase, identified in museum register as black and white syenite, perhaps hornblende diorite (?), barrel-shaped type 39. From Naqada Tomb T 16
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 50, Interior 14
Scan note	Scan only partly capture the vessel

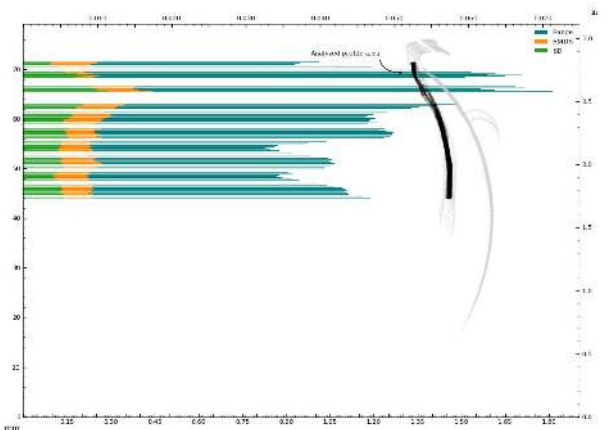
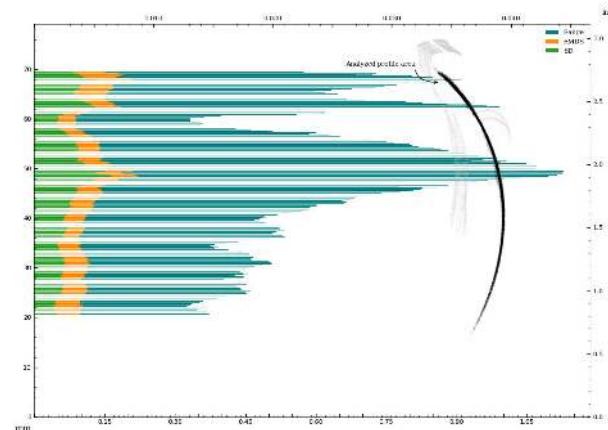
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0199	0.141	0.091	0.050	1.298	−0.781	0.516	170208
Interior	0.6061	0.779	0.385	0.312	3.145	−1.529	1.616	43592
Interior separate	0.0699	0.264	0.154	0.097	2.286	−0.944	1.342	43220
	in ²	in	in	in	in	in	in	
Exterior	0.000031	0.0056	0.0036	0.0020	0.0511	−0.0308	0.0203	170208
Interior	0.000939	0.0306	0.0151	0.0123	0.1238	−0.0602	0.0636	43592
Interior separate	0.000108	0.0104	0.0061	0.0038	0.0900	−0.0372	0.0528	43220

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.572	0.326	1.128	0.071	0.041	0.165	0.117	0.084	0.221	244	0.200
Interior	1.114	0.824	1.813	0.135	0.092	0.288	0.243	0.210	0.443	119	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0225	0.0128	0.0444	0.0028	0.0016	0.0065	0.0046	0.0033	0.0087	244	0.0079
Interior	0.0439	0.0324	0.0714	0.0053	0.0036	0.0113	0.0096	0.0083	0.0174	119	0.0079



MV015c



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Artifact Precision Report, version 01.20	
Material	Black & White Syenite, perhaps Hornblende Diorite
Description	Stone vase, identified in museum register as black and white syenite, perhaps hornblende diorite (?), barrel-shaped type 39. From Naqada Tomb T 16
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 52, Interior 12

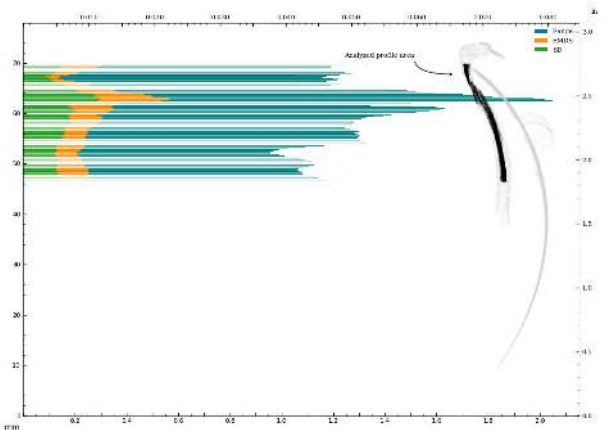
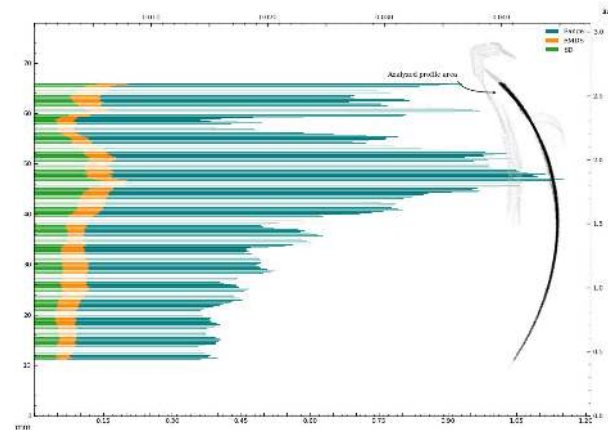
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0194	0.139	0.090	0.049	1.283	−0.765	0.518	496062
Interior	0.5218	0.722	0.387	0.305	3.088	−1.547	1.542	140033
Interior separate	0.0812	0.285	0.177	0.107	2.175	−1.168	1.007	137576
	in ²	in	in	in	in	in	in	
Exterior	0.000030	0.0055	0.0035	0.0019	0.0505	−0.0301	0.0204	496062
Interior	0.000809	0.0284	0.0152	0.0120	0.1216	−0.0609	0.0607	140033
Interior separate	0.000126	0.0112	0.0070	0.0042	0.0856	−0.0460	0.0396	137576

Circularity measurements

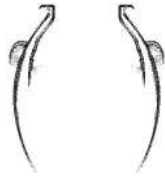
Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.564	0.345	1.152	0.068	0.044	0.161	0.117	0.070	0.215	275	0.200
Interior	1.243	0.942	2.050	0.152	0.086	0.298	0.254	0.131	0.566	118	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0222	0.0136	0.0454	0.0027	0.0017	0.0063	0.0046	0.0028	0.0085	275	0.0079
Interior	0.0489	0.0371	0.0807	0.0060	0.0034	0.0117	0.0100	0.0052	0.0223	118	0.0079



MV016b



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Artifact Precision Report, version 01.20

Material	Basalt
Description	Stone vase, basalt, barrel shaped, type 55. From Naqada Tomb T 16
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 2.12, Interior 12

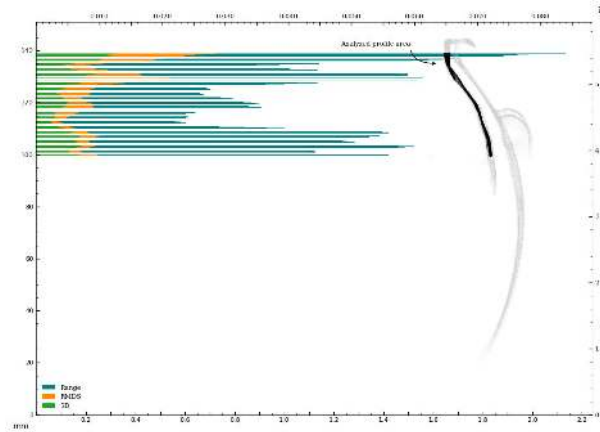
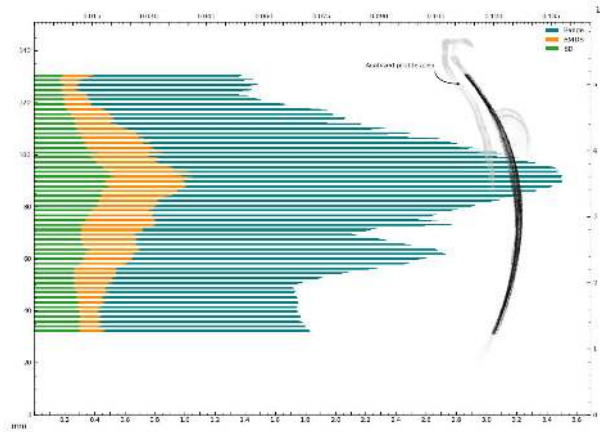
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.4711	0.686	0.394	0.294	3.705	−2.251	1.454	221522
Interior	0.6029	0.776	0.360	0.255	3.656	−1.965	1.691	97118
Interior separate	0.0853	0.292	0.202	0.096	2.307	−1.059	1.248	97211
	in ²	in	in	in	in	in	in	
Exterior	0.000730	0.0270	0.0155	0.0116	0.1458	−0.0886	0.0572	221522
Interior	0.000935	0.0306	0.0142	0.0100	0.1439	−0.0774	0.0666	97118
Interior separate	0.000132	0.0115	0.0080	0.0038	0.0908	−0.0417	0.0491	97211

Circularity measurements

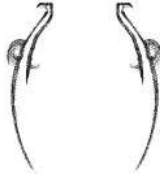
Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	2.287	1.308	3.515	0.305	0.155	0.533	0.620	0.253	1.056	500	0.200
Interior	1.061	0.550	2.134	0.133	0.057	0.317	0.222	0.089	0.724	197	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0900	0.0515	0.1384	0.0120	0.0061	0.0210	0.0244	0.0099	0.0416	500	0.0079
Interior	0.0418	0.0216	0.0840	0.0052	0.0023	0.0125	0.0088	0.0035	0.0285	197	0.0079



MV016c



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Artifact Precision Report, version 01.20	
Material	Basalt
Description	Stone vase, basalt, barrel shaped, type 55. From Naqada Tomb T 16
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 2.27, Interior 2.76

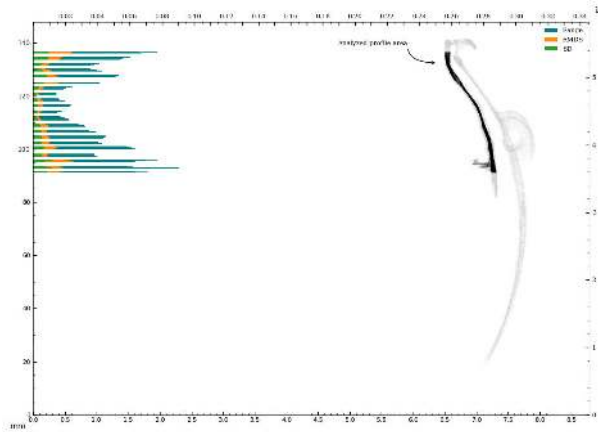
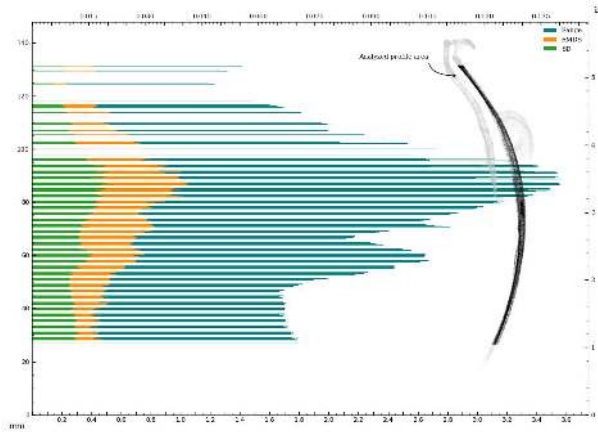
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.4399	0.663	0.387	0.277	3.752	−2.277	1.475	81197
Interior	0.9321	0.965	0.635	0.297	9.040	−6.787	2.253	76725
Interior separate	0.3620	0.602	0.514	0.120	8.044	−6.384	1.661	77028
	in ²	in	in	in	in	in	in	
Exterior	0.000682	0.0261	0.0152	0.0109	0.1477	−0.0896	0.0581	81197
Interior	0.001445	0.0380	0.0250	0.0117	0.3559	−0.2672	0.0887	76725
Interior separate	0.000561	0.0237	0.0202	0.0047	0.3167	−0.2513	0.0654	77028

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	2.343	1.185	3.571	0.318	0.132	0.548	0.672	0.215	1.063	405	0.200
Interior	1.009	0.362	8.364	0.129	0.046	1.341	0.241	0.081	2.168	221	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0922	0.0467	0.1406	0.0125	0.0052	0.0216	0.0264	0.0085	0.0418	405	0.0079
Interior	0.0397	0.0142	0.3293	0.0051	0.0018	0.0528	0.0095	0.0032	0.0853	221	0.0079



MV018b



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Artifact Precision Report, version 01.20	
Material	Diorite
Description	Vase of diorite, squat, type 15, rim and one handle broken. From Naqada Cemetery T Tomb 20
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 13, Interior 65

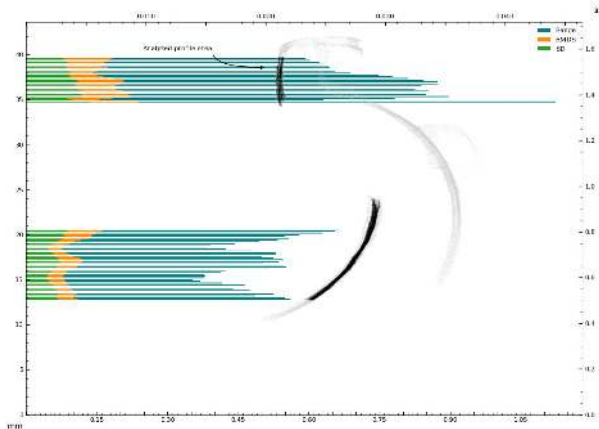
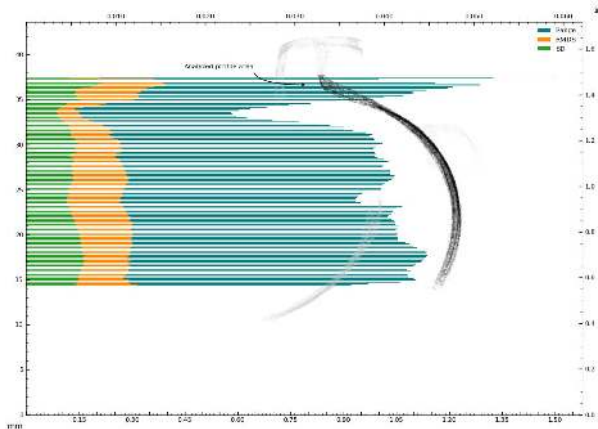
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0764	0.276	0.150	0.107	3.484	−0.826	2.658	255111
Interior	0.0770	0.278	0.162	0.127	1.709	−0.827	0.883	148411
Interior separate	0.0154	0.124	0.079	0.048	0.956	−0.433	0.523	148122
	in ²	in	in	in	in	in	in	
Exterior	0.000118	0.0109	0.0059	0.0042	0.1371	−0.0325	0.1046	255111
Interior	0.000119	0.0109	0.0064	0.0050	0.0673	−0.0326	0.0347	148411
Interior separate	0.000024	0.0049	0.0031	0.0019	0.0376	−0.0170	0.0206	148122

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.027	0.580	1.502	0.136	0.088	0.214	0.282	0.137	0.398	119	0.200
Interior	0.550	0.353	1.124	0.072	0.045	0.135	0.119	0.077	0.239	65	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0405	0.0228	0.0591	0.0053	0.0035	0.0084	0.0111	0.0054	0.0157	119	0.0079
Interior	0.0216	0.0139	0.0443	0.0029	0.0018	0.0053	0.0047	0.0030	0.0094	65	0.0079



MV019b



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Artifact Precision Report, version 01.20

Material	Porphyry
Description	Stone vase, type 99, porphyry, rim damaged. From Naqada Cemetery B Tomb 99
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 15, Interior 31

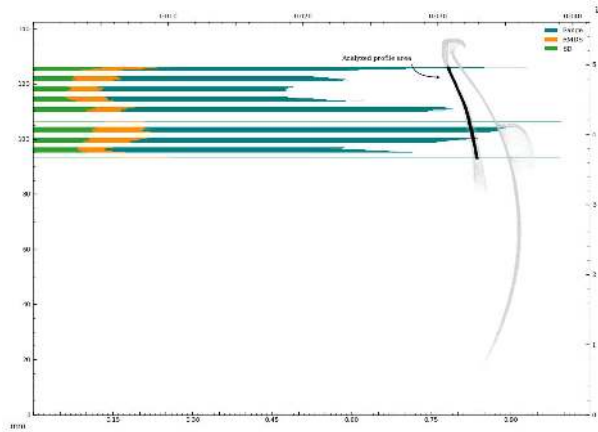
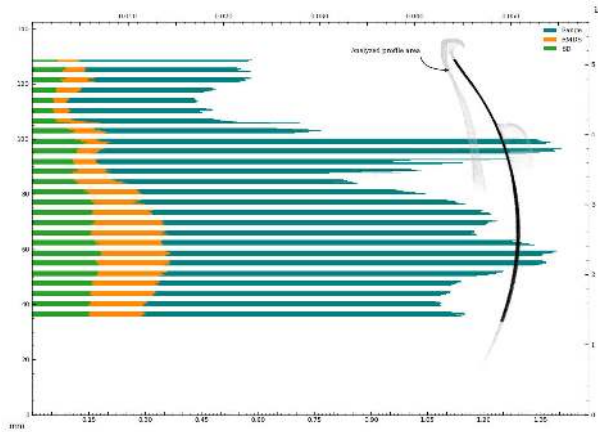
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0649	0.255	0.156	0.101	1.608	−0.699	0.909	226479
Interior	0.5854	0.765	0.404	0.300	3.119	−1.595	1.524	58717
Interior separate	0.0327	0.181	0.105	0.070	1.091	−0.509	0.582	60307
	in ²	in	in	in	in	in	in	
Exterior	0.000101	0.0100	0.0061	0.0040	0.0633	−0.0275	0.0358	226479
Interior	0.000907	0.0301	0.0159	0.0118	0.1228	−0.0628	0.0600	58717
Interior separate	0.000051	0.0071	0.0041	0.0028	0.0430	−0.0201	0.0229	60307

Circularity measurements

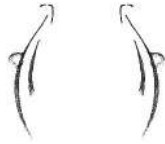
Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.096	0.421	1.406	0.144	0.049	0.185	0.270	0.090	0.367	475	0.200
Interior	0.718	0.473	0.998	0.101	0.056	0.162	0.170	0.122	0.256	167	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0431	0.0166	0.0554	0.0057	0.0019	0.0073	0.0106	0.0035	0.0144	475	0.0079
Interior	0.0283	0.0186	0.0393	0.0040	0.0022	0.0064	0.0067	0.0048	0.0101	167	0.0079



MV020b



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Artifact Precision Report, version 01.20	
Material	Basalt
Description	Stone vase, basalt, type B 37. from Naqada Tomb 100.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 7, Interior 180

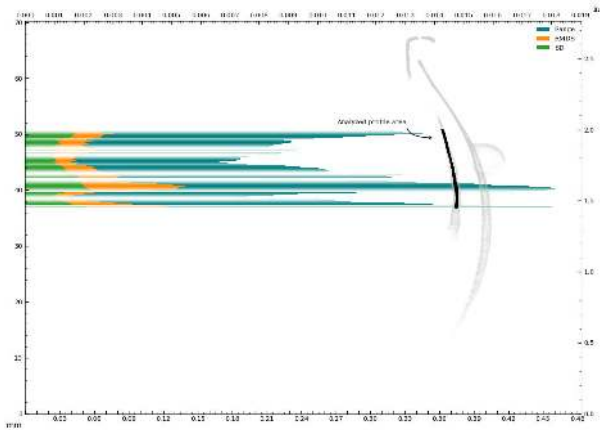
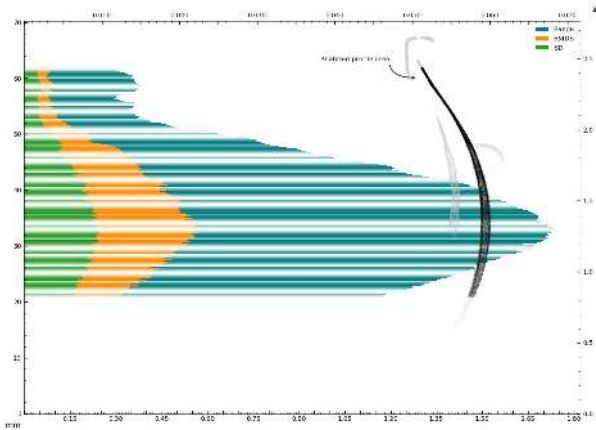
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

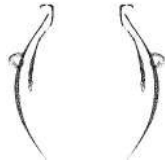
Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.1389	0.373	0.226	0.177	1.808	−0.870	0.938	70440
Interior	0.1345	0.367	0.189	0.165	1.406	−0.677	0.730	24610
Interior separate	0.0056	0.075	0.044	0.030	0.480	−0.248	0.232	24370
	in ²	in	in	in	in	in	in	
Exterior	0.000215	0.0147	0.0089	0.0070	0.0712	−0.0342	0.0369	70440
Interior	0.000209	0.0144	0.0075	0.0065	0.0554	−0.0266	0.0287	24610
Interior separate	0.000009	0.0029	0.0017	0.0012	0.0189	−0.0098	0.0091	24370

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.256	0.295	1.734	0.171	0.038	0.248	0.375	0.072	0.585	205	0.200
Interior	0.261	0.169	0.460	0.034	0.021	0.056	0.061	0.035	0.139	70	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0494	0.0116	0.0683	0.0067	0.0015	0.0098	0.0148	0.0028	0.0230	205	0.0079
Interior	0.0103	0.0066	0.0181	0.0013	0.0008	0.0022	0.0024	0.0014	0.0055	70	0.0079



MV020c



Artifact Precision Report, version 01.20	
Material	Basalt
Description	Stone vase, basalt, type B 37. from Naqada Tomb 100.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 8, Interior 97

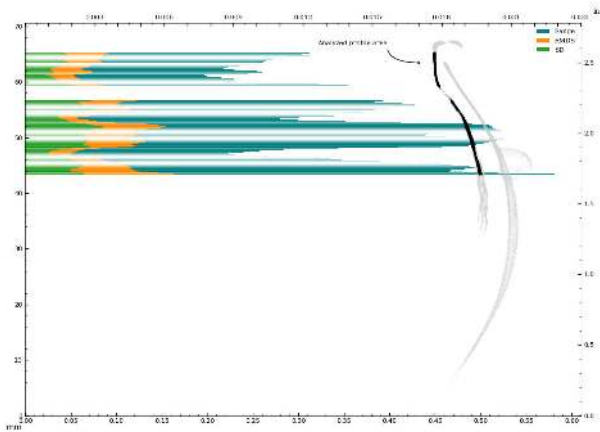
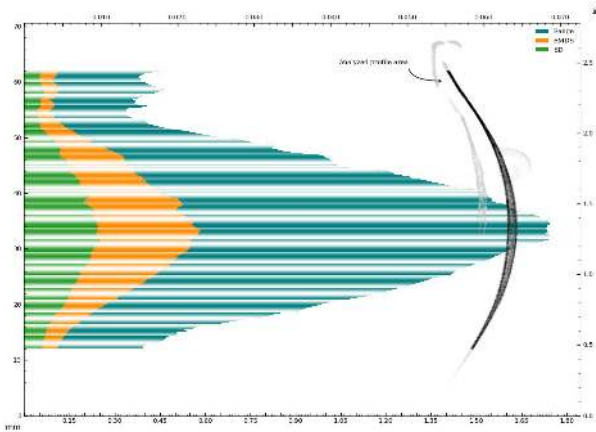
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.1326	0.364	0.221	0.155	1.822	−0.894	0.928	211714
Interior	0.0512	0.226	0.140	0.075	1.398	−0.801	0.597	84644
Interior separate	0.0103	0.102	0.060	0.039	0.607	−0.320	0.287	83396
	in ²	in	in	in	in	in	in	
Exterior	0.000206	0.0143	0.0087	0.0061	0.0717	−0.0352	0.0366	211714
Interior	0.000079	0.0089	0.0055	0.0030	0.0550	−0.0315	0.0235	84644
Interior separate	0.000016	0.0040	0.0024	0.0015	0.0239	−0.0126	0.0113	83396

Circularity measurements

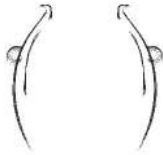
Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.018	0.334	1.756	0.142	0.040	0.250	0.328	0.085	0.585	250	0.200
Interior	0.357	0.196	0.581	0.051	0.023	0.085	0.093	0.051	0.163	98	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0401	0.0132	0.0691	0.0056	0.0016	0.0098	0.0129	0.0033	0.0230	250	0.0079
Interior	0.0140	0.0077	0.0229	0.0020	0.0009	0.0034	0.0036	0.0020	0.0064	98	0.0079



MV021b



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Artifact Precision Report, version 01.20	
Material	Grey Basalt
Description	Grey basalt vase with two perforated handles. From Naqada Tomb 1207
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 9, Interior 66

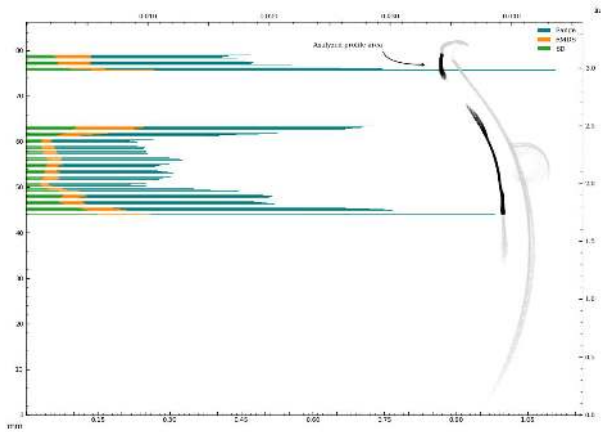
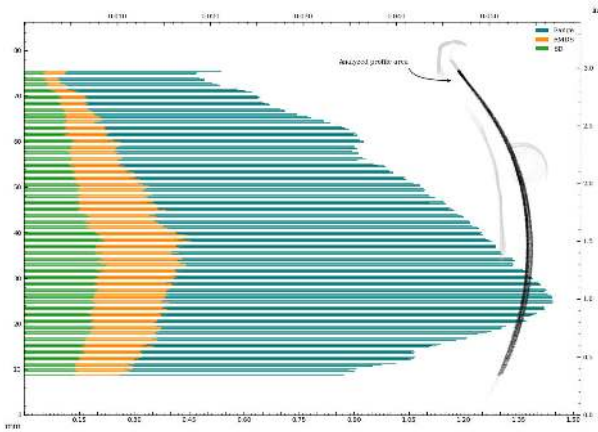
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

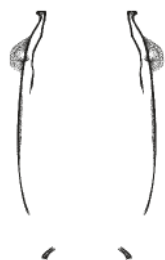
Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.1072	0.327	0.180	0.141	1.500	−0.706	0.795	126102
Interior	0.0280	0.167	0.091	0.064	1.243	−0.516	0.727	45843
Interior separate	0.0151	0.123	0.078	0.048	0.953	−0.405	0.549	45850
	in ²	in	in	in	in	in	in	
Exterior	0.000166	0.0129	0.0071	0.0056	0.0591	−0.0278	0.0313	126102
Interior	0.000043	0.0066	0.0036	0.0025	0.0489	−0.0203	0.0286	45843
Interior separate	0.000023	0.0048	0.0031	0.0019	0.0375	−0.0159	0.0216	45850

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.104	0.458	1.446	0.151	0.050	0.227	0.337	0.089	0.455	335	0.200
Interior	0.371	0.216	1.109	0.049	0.029	0.149	0.085	0.046	0.267	117	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0435	0.0180	0.0569	0.0059	0.0020	0.0089	0.0133	0.0035	0.0179	335	0.0079
Interior	0.0146	0.0085	0.0437	0.0019	0.0011	0.0059	0.0034	0.0018	0.0105	117	0.0079



MV022



Artifact Precision Report, version 01.20

Material Basalt

Description Stone vase, basalt, footed type T119 edge badly chipped. From Naqada Tomb 1782.

Collection Petrie Museum of Egyptian Archaeology

Provenance Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie

Precision Score Exterior 7, Interior 37

Scan note Damaged part of the vessel body have been removed.

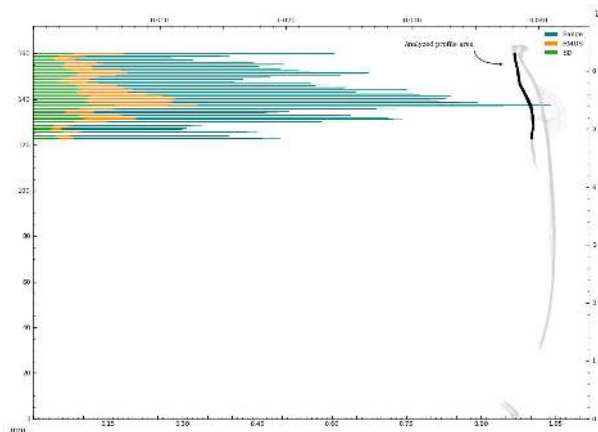
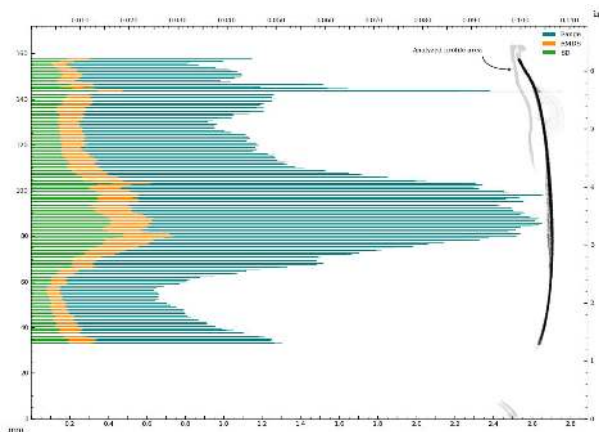
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.1496	0.387	0.265	0.135	4.385	−1.961	2.423	258972
Interior	0.0646	0.254	0.140	0.102	1.352	−0.682	0.669	128123
Interior separate	0.0268	0.164	0.101	0.065	1.065	−0.428	0.637	128041
	in ²	in	in	in	in	in	in	
Exterior	0.000232	0.0152	0.0104	0.0053	0.1726	−0.0772	0.0954	258972
Interior	0.000100	0.0100	0.0055	0.0040	0.0532	−0.0269	0.0263	128123
Interior separate	0.000042	0.0064	0.0040	0.0026	0.0419	−0.0169	0.0251	128041

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.210	0.626	2.750	0.164	0.078	0.507	0.291	0.147	0.787	627	0.200
Interior	0.546	0.277	1.065	0.074	0.033	0.163	0.144	0.057	0.343	190	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0476	0.0246	0.1083	0.0065	0.0031	0.0200	0.0115	0.0058	0.0310	627	0.0079
Interior	0.0215	0.0109	0.0419	0.0029	0.0013	0.0064	0.0057	0.0023	0.0135	190	0.0079



MV023



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Artifact Precision Report, version 01.20	
Material	Basalt
Description	Tapering barrel vase in basalt, with lug handles perforated vertically.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 21, Interior 39

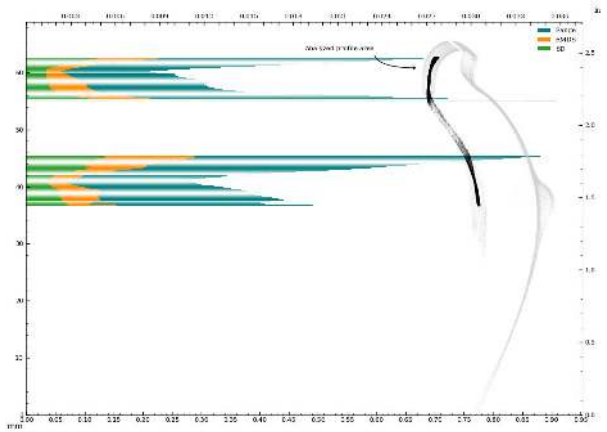
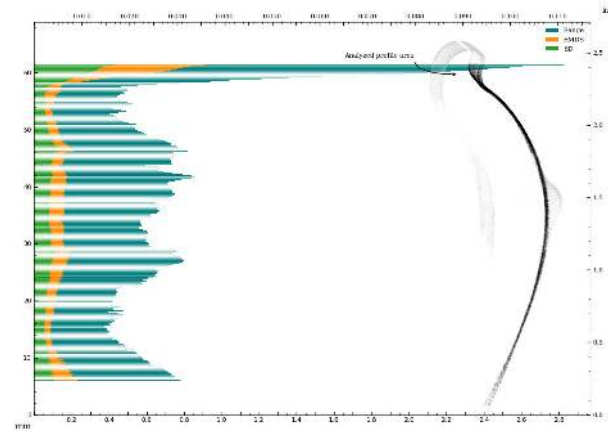
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0478	0.219	0.161	0.061	2.732	−1.160	1.572	144965
Interior	0.5383	0.734	0.359	0.293	2.723	−1.457	1.266	34269
Interior separate	0.0256	0.160	0.100	0.052	1.219	−0.673	0.545	34613
	in ²	in	in	in	in	in	in	
Exterior	0.000074	0.0086	0.0063	0.0024	0.1076	−0.0457	0.0619	144965
Interior	0.000834	0.0289	0.0141	0.0116	0.1072	−0.0574	0.0498	34269
Interior separate	0.000040	0.0063	0.0039	0.0020	0.0480	−0.0265	0.0215	34613

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.604	0.375	2.827	0.081	0.051	0.423	0.151	0.081	0.913	277	0.200
Interior	0.393	0.213	0.907	0.058	0.033	0.142	0.124	0.060	0.301	83	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0238	0.0148	0.1113	0.0032	0.0020	0.0167	0.0060	0.0032	0.0359	277	0.0079
Interior	0.0155	0.0084	0.0357	0.0023	0.0013	0.0056	0.0049	0.0024	0.0119	83	0.0079



MV023b



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Artifact Precision Report, version 01.20	
Material	Basalt
Description	Tapering barrel vase in basalt, with lug handles perforated vertically.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 17, Interior 28

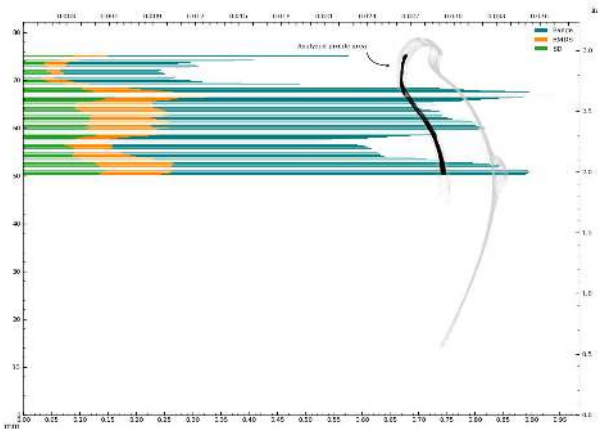
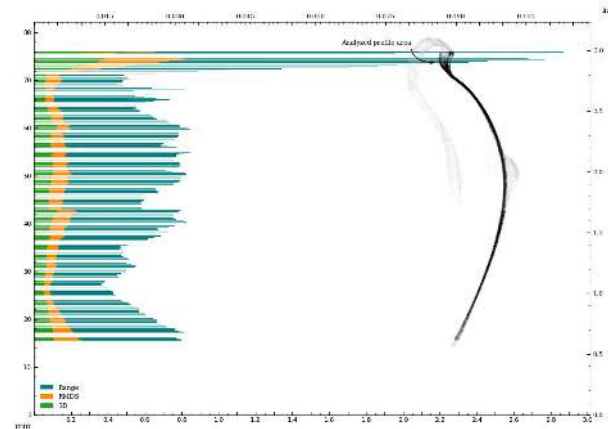
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0588	0.242	0.177	0.066	2.896	−1.504	1.392	392683
Interior	0.4296	0.655	0.328	0.269	2.621	−1.422	1.199	78074
Interior separate	0.0353	0.188	0.115	0.077	1.059	−0.487	0.572	78643
	in ²	in	in	in	in	in	in	
Exterior	0.000091	0.0095	0.0070	0.0026	0.1140	−0.0592	0.0548	392683
Interior	0.000666	0.0258	0.0129	0.0106	0.1032	−0.0560	0.0472	78074
Interior separate	0.000055	0.0074	0.0045	0.0030	0.0417	−0.0192	0.0225	78643

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.662	0.362	2.871	0.083	0.049	0.489	0.159	0.083	0.822	305	0.200
Interior	0.709	0.240	0.938	0.098	0.031	0.164	0.227	0.057	0.332	125	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0261	0.0143	0.1130	0.0033	0.0019	0.0192	0.0063	0.0033	0.0324	305	0.0079
Interior	0.0279	0.0094	0.0369	0.0039	0.0012	0.0065	0.0089	0.0022	0.0131	125	0.0079



MV024



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Artifact Precision Report, version 01.20	
Material	Basalt
Description	Squat basalt vase with two cylindrical handles (Petrie type 3). Rim damaged
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 1.84, Interior 14

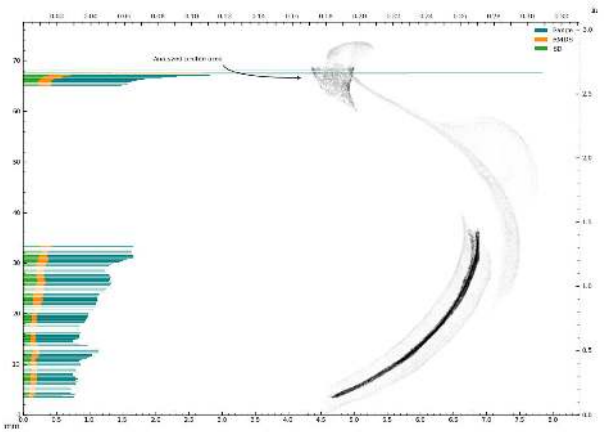
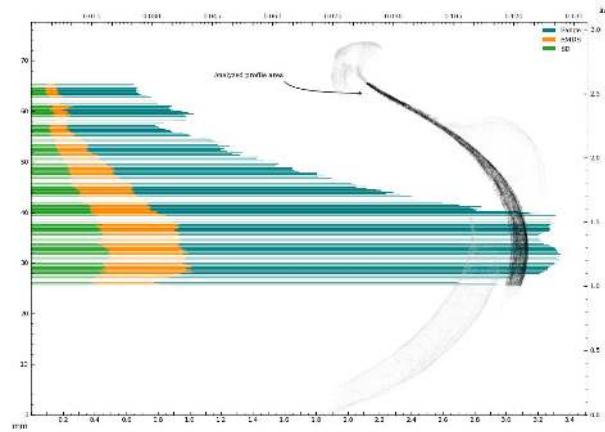
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.5433	0.737	0.443	0.341	3.405	−1.982	1.423	140737
Interior	5.4975	2.345	1.057	0.889	9.081	−5.211	3.870	212249
Interior separate	0.0703	0.265	0.157	0.100	1.703	−1.059	0.644	222773
	in ²	in	in	in	in	in	in	
Exterior	0.000842	0.0290	0.0175	0.0134	0.1341	−0.0781	0.0560	140737
Interior	0.008521	0.0923	0.0416	0.0350	0.3575	−0.2051	0.1524	212249
Interior separate	0.000109	0.0104	0.0062	0.0039	0.0671	−0.0417	0.0254	222773

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	2.049	0.639	3.345	0.255	0.086	0.507	0.629	0.143	1.017	200	0.200
Interior	1.086	0.588	8.003	0.147	0.073	1.343	0.252	0.153	2.749	170	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0807	0.0252	0.1317	0.0100	0.0034	0.0199	0.0248	0.0056	0.0401	200	0.0079
Interior	0.0428	0.0231	0.3151	0.0058	0.0029	0.0529	0.0099	0.0060	0.1082	170	0.0079



MV024b



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Artifact Precision Report, version 01.20

Material	Basalt
Description	Squat basalt vase with two cylindrical handles (Petrie type 3). Rim damaged
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 1.86, Interior 18

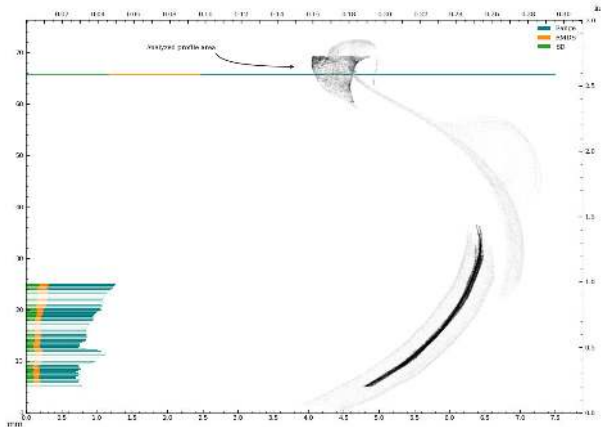
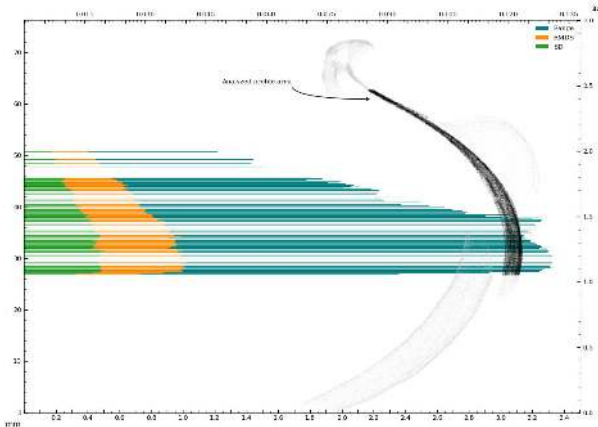
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.5389	0.734	0.436	0.337	3.425	−1.980	1.445	39295
Interior	5.3510	2.313	1.042	0.869	7.423	−3.758	3.665	53480
Interior separate	0.0542	0.233	0.133	0.091	1.373	−0.772	0.601	53823
	in ²	in	in	in	in	in	in	
Exterior	0.000835	0.0289	0.0171	0.0133	0.1348	−0.0780	0.0569	39295
Interior	0.008294	0.0911	0.0410	0.0342	0.2922	−0.1479	0.1443	53480
Interior separate	0.000084	0.0092	0.0052	0.0036	0.0541	−0.0304	0.0237	53823

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	3.161	1.220	3.335	0.409	0.172	0.508	0.879	0.401	1.031	99	0.200
Interior	0.946	0.697	7.508	0.122	0.091	1.177	0.212	0.177	2.468	103	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.1244	0.0480	0.1313	0.0161	0.0068	0.0200	0.0346	0.0158	0.0406	99	0.0079
Interior	0.0372	0.0274	0.2956	0.0048	0.0036	0.0463	0.0084	0.0070	0.0971	103	0.0079



MV025



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Artifact Precision Report, version 01.20

Material	Basalt
Description	Squat basalt vase, with two cylindrical handles (Petrie type 5). Rim damaged.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 4, Interior 14

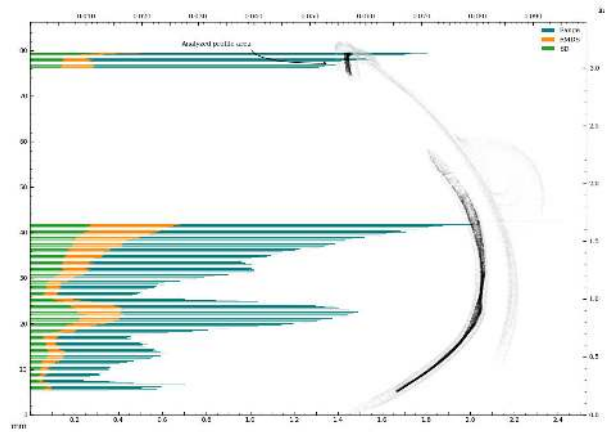
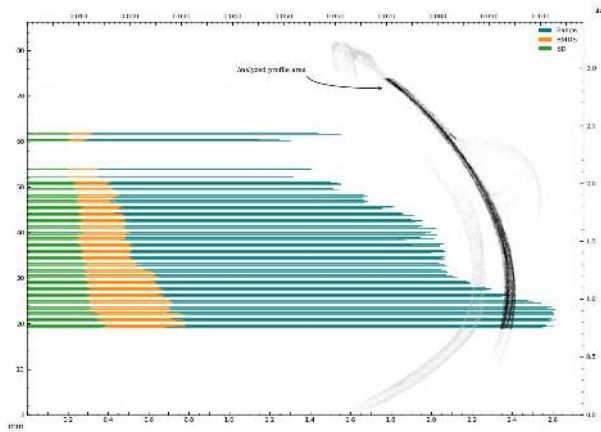
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.2584	0.508	0.304	0.227	2.749	−1.277	1.472	48867
Interior	0.9890	0.994	0.470	0.323	5.605	−3.615	1.990	68593
Interior separate	0.0713	0.267	0.187	0.086	2.136	−1.094	1.043	66426
	in ²	in	in	in	in	in	in	
Exterior	0.000400	0.0200	0.0120	0.0089	0.1082	−0.0503	0.0580	48867
Interior	0.001533	0.0392	0.0185	0.0127	0.2207	−0.1423	0.0783	68593
Interior separate	0.000111	0.0105	0.0073	0.0034	0.0841	−0.0431	0.0410	66426

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	2.041	0.961	2.621	0.273	0.120	0.392	0.510	0.214	0.787	175	0.200
Interior	0.956	0.242	2.408	0.139	0.032	0.345	0.253	0.053	0.776	202	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0804	0.0378	0.1032	0.0107	0.0047	0.0154	0.0201	0.0084	0.0310	175	0.0079
Interior	0.0376	0.0095	0.0948	0.0055	0.0013	0.0136	0.0100	0.0021	0.0305	202	0.0079



MV025b



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Artifact Precision Report, version 01.20	
Material	Basalt
Description	Squat basalt vase, with two cylindrical handles (Petrie type 5). Rim damaged.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 3, Interior 13

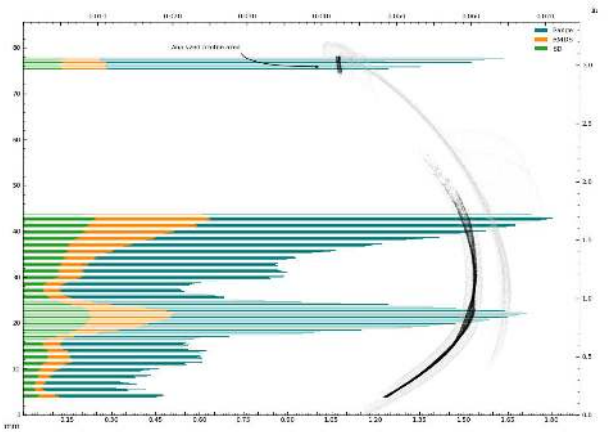
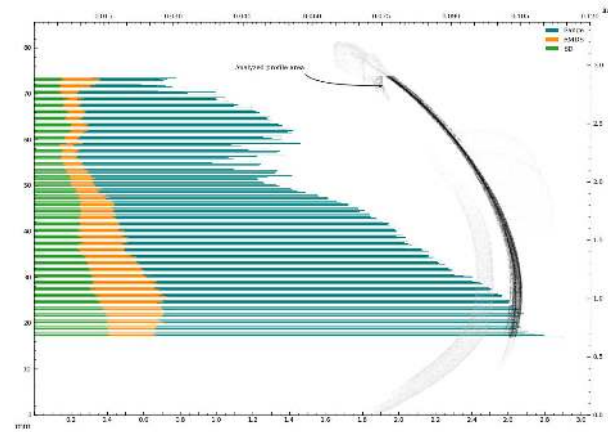
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.3040	0.551	0.379	0.211	5.206	−4.103	1.104	112582
Interior	1.8308	1.353	0.653	0.519	8.403	−4.594	3.809	125361
Interior separate	0.0749	0.274	0.187	0.088	1.979	−0.872	1.107	126961
	in ²	in	in	in	in	in	in	
Exterior	0.000471	0.0217	0.0149	0.0083	0.2050	−0.1615	0.0435	112582
Interior	0.002838	0.0533	0.0257	0.0204	0.3308	−0.1809	0.1500	125361
Interior separate	0.000116	0.0108	0.0074	0.0035	0.0779	−0.0343	0.0436	126961

Circularity measurements

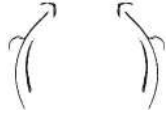
Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.753	0.559	2.906	0.245	0.118	0.422	0.434	0.212	0.731	285	0.200
Interior	0.883	0.288	1.803	0.122	0.036	0.248	0.219	0.062	0.641	215	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0690	0.0220	0.1144	0.0096	0.0046	0.0166	0.0171	0.0084	0.0288	285	0.0079
Interior	0.0348	0.0114	0.0710	0.0048	0.0014	0.0098	0.0086	0.0024	0.0252	215	0.0079



MV026



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Artifact Precision Report, version 01.20	
Material	Basalt
Description	Basalt barrel vase, with two cylindrical handles and flat base.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 39, Interior 322

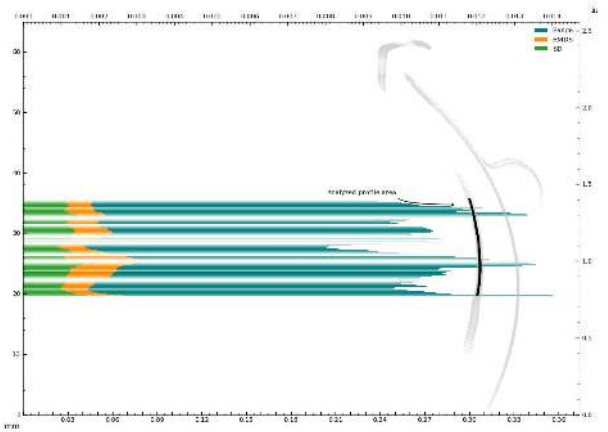
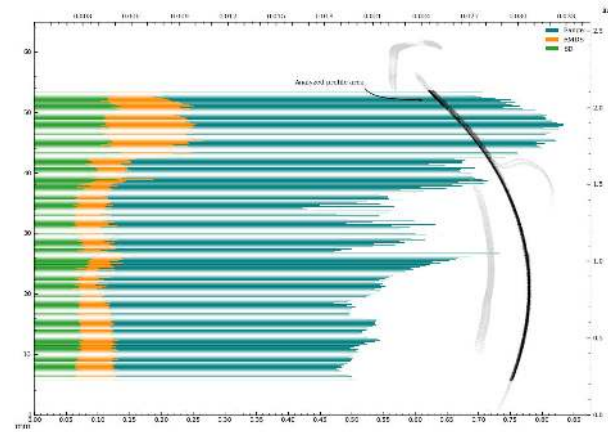
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0255	0.160	0.099	0.062	0.884	−0.343	0.541	75736
Interior	0.0981	0.313	0.153	0.144	1.141	−0.557	0.584	46948
Interior separate	0.0031	0.056	0.032	0.021	0.378	−0.206	0.172	46919
	in ²	in	in	in	in	in	in	
Exterior	0.000040	0.0063	0.0039	0.0025	0.0348	−0.0135	0.0213	75736
Interior	0.000152	0.0123	0.0060	0.0057	0.0449	−0.0219	0.0230	46948
Interior separate	0.000005	0.0022	0.0013	0.0008	0.0149	−0.0081	0.0068	46919

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.574	0.423	0.834	0.073	0.058	0.144	0.127	0.096	0.265	240	0.200
Interior	0.269	0.204	0.356	0.030	0.023	0.043	0.053	0.041	0.078	80	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0226	0.0166	0.0329	0.0029	0.0023	0.0057	0.0050	0.0038	0.0104	240	0.0079
Interior	0.0106	0.0080	0.0140	0.0012	0.0009	0.0017	0.0021	0.0016	0.0031	80	0.0079



MV027



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Artifact Precision Report, version 01.20	
Material	Basalt
Description	Basalt barrel vase, with two lug handles perforated horizontally. Flat base, rim ground away.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 7, Interior 234

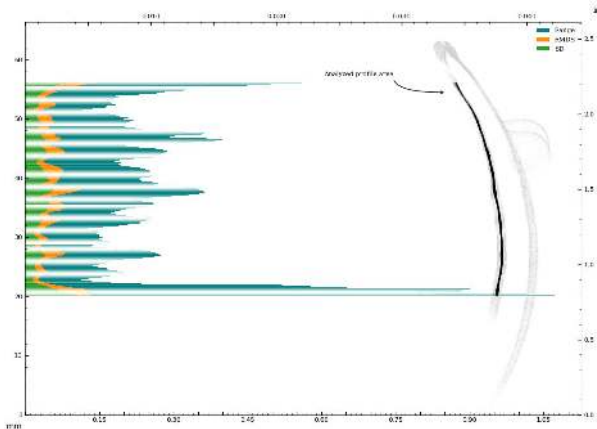
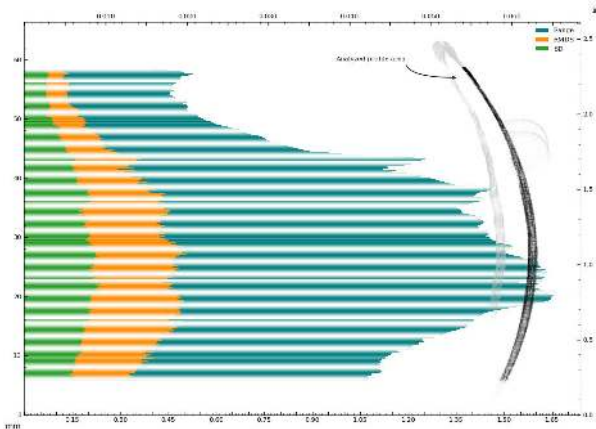
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.1535	0.392	0.210	0.178	1.752	−0.811	0.941	101132
Interior	0.1750	0.418	0.215	0.174	2.140	−1.312	0.828	110477
Interior separate	0.0043	0.065	0.042	0.025	1.151	−0.950	0.201	110347
	in ²	in	in	in	in	in	in	
Exterior	0.000238	0.0154	0.0083	0.0070	0.0690	−0.0319	0.0370	101132
Interior	0.000271	0.0165	0.0085	0.0068	0.0843	−0.0517	0.0326	110477
Interior separate	0.000007	0.0026	0.0016	0.0010	0.0453	−0.0374	0.0079	110347

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.285	0.404	1.655	0.178	0.054	0.239	0.408	0.100	0.518	265	0.200
Interior	0.230	0.114	1.072	0.032	0.012	0.102	0.057	0.021	0.144	179	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0506	0.0159	0.0652	0.0070	0.0021	0.0094	0.0161	0.0039	0.0204	265	0.0079
Interior	0.0091	0.0045	0.0422	0.0012	0.0005	0.0040	0.0022	0.0008	0.0057	179	0.0079



MV027b



Petrie Museum, CC BY-NC-SA



Artifact Precision Report, version 01.20	
Material	Basalt
Description	Basalt barrel vase, with two lug handles perforated horizontally. Flat base, rim ground away.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 8, Interior 223

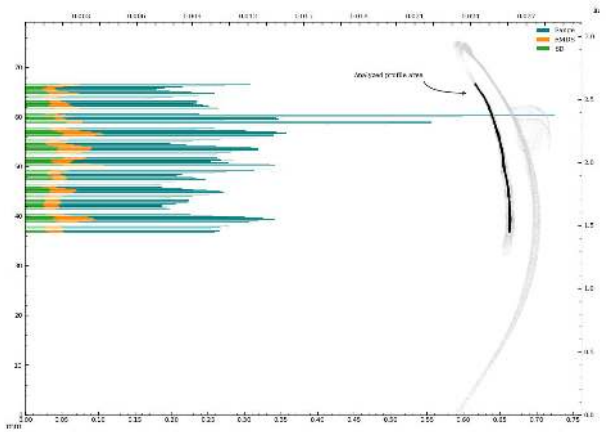
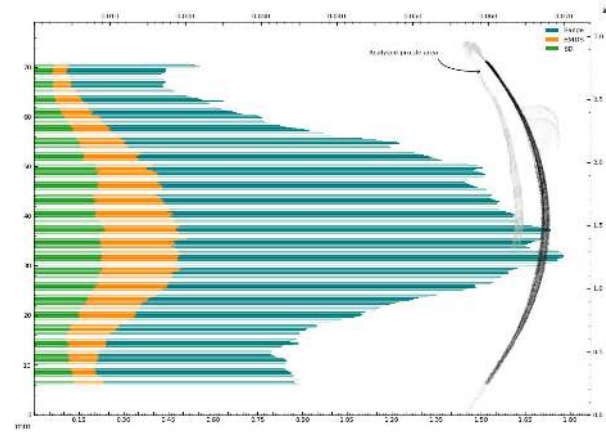
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.1298	0.360	0.206	0.147	1.852	−0.954	0.898	380466
Interior	0.1890	0.435	0.228	0.172	1.889	−1.008	0.881	235281
Interior separate	0.0045	0.067	0.040	0.027	0.815	−0.606	0.208	234153
	in ²	in	in	in	in	in	in	
Exterior	0.000201	0.0142	0.0081	0.0058	0.0729	−0.0376	0.0354	380466
Interior	0.000293	0.0171	0.0090	0.0068	0.0744	−0.0397	0.0347	235281
Interior separate	0.000007	0.0026	0.0016	0.0011	0.0321	−0.0239	0.0082	234153

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.196	0.335	1.778	0.153	0.044	0.237	0.342	0.085	0.518	325	0.200
Interior	0.258	0.164	0.725	0.034	0.022	0.056	0.059	0.038	0.116	150	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0471	0.0132	0.0700	0.0060	0.0017	0.0093	0.0135	0.0034	0.0204	325	0.0079
Interior	0.0102	0.0065	0.0286	0.0013	0.0009	0.0022	0.0023	0.0015	0.0046	150	0.0079



MV028



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Artifact Precision Report, version 01.20	
Material	Basalt
Description	Basalt barrel vase with two cylindrical handles and flat base. Rim chipped.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 10, Interior 75

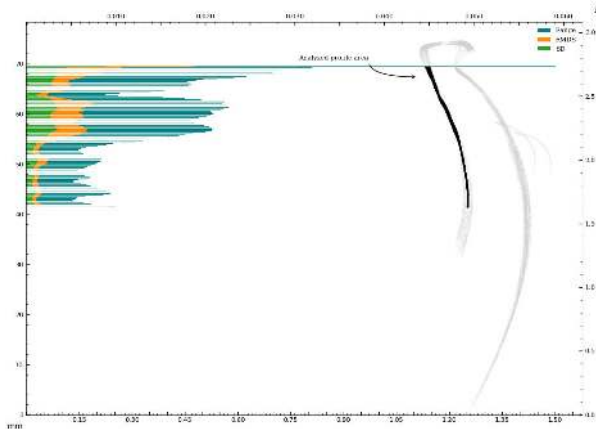
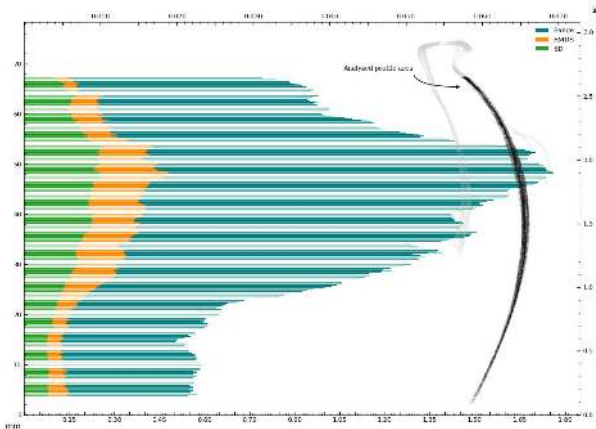
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.1051	0.324	0.207	0.141	1.764	−0.972	0.792	215178
Interior	0.1519	0.390	0.228	0.146	2.003	−0.983	1.020	55699
Interior separate	0.0133	0.115	0.080	0.038	0.899	−0.474	0.424	55677
	in ²	in	in	in	in	in	in	
Exterior	0.000163	0.0128	0.0082	0.0056	0.0695	−0.0383	0.0312	215178
Interior	0.000235	0.0153	0.0090	0.0058	0.0789	−0.0387	0.0401	55699
Interior separate	0.000021	0.0045	0.0031	0.0015	0.0354	−0.0187	0.0167	55677

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.104	0.504	1.762	0.159	0.068	0.264	0.268	0.117	0.485	318	0.200
Interior	0.322	0.132	1.502	0.039	0.016	0.223	0.080	0.029	0.477	142	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0435	0.0198	0.0694	0.0063	0.0027	0.0104	0.0106	0.0046	0.0191	318	0.0079
Interior	0.0127	0.0052	0.0591	0.0016	0.0006	0.0088	0.0032	0.0011	0.0188	142	0.0079



MV029



Petrie Museum, CC BY-NC-SA



Artifact Precision Report, version 01.20	
Material	Basalt
Description	Basalt tubular vase, with two horizontally perforated lug handles
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 6, Interior 32

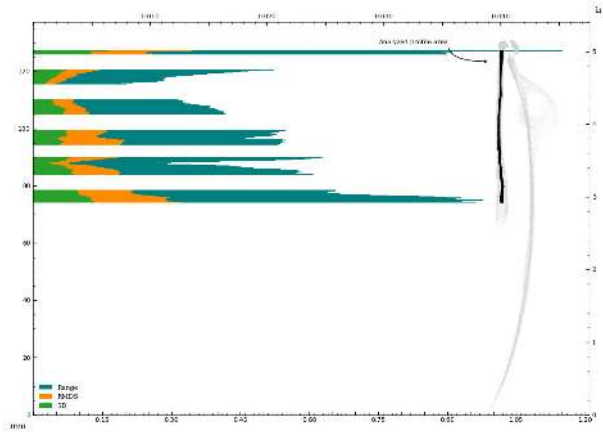
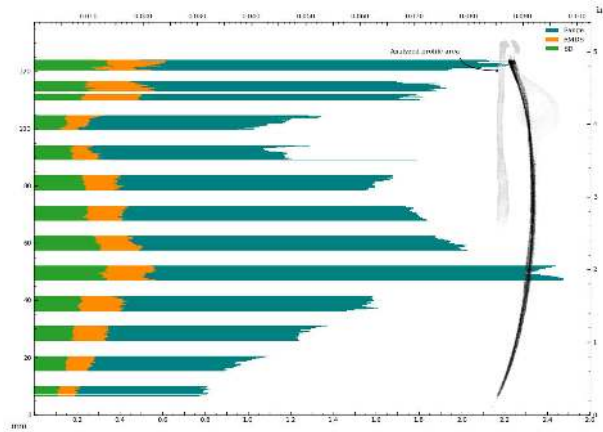
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.1773	0.421	0.260	0.175	2.944	−1.430	1.513	252985
Interior	0.6186	0.787	0.394	0.232	3.890	−1.797	2.093	152361
Interior separate	0.0317	0.178	0.100	0.081	1.051	−0.497	0.554	151354
	in ²	in	in	in	in	in	in	
Exterior	0.000275	0.0166	0.0102	0.0069	0.1159	−0.0563	0.0596	252985
Interior	0.000959	0.0310	0.0155	0.0091	0.1531	−0.0708	0.0824	152361
Interior separate	0.000049	0.0070	0.0040	0.0032	0.0414	−0.0196	0.0218	151354

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.622	0.771	2.478	0.227	0.105	0.375	0.403	0.188	0.628	588	0.200
Interior	0.537	0.192	1.151	0.074	0.024	0.164	0.158	0.045	0.347	267	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0638	0.0304	0.0976	0.0089	0.0041	0.0148	0.0159	0.0074	0.0247	588	0.0079
Interior	0.0211	0.0076	0.0453	0.0029	0.0009	0.0065	0.0062	0.0018	0.0136	267	0.0079



MV030



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Artifact Precision Report, version 01.20

Material	Basalt
Description	Basalt tubular vase, with two horizontally perforated lug handles, with double lozenge or mouth sign below shoulder midway between lug handles on one side. Underside marked 1868, so possibly from Naqada tomb 1868.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 26, Interior 87

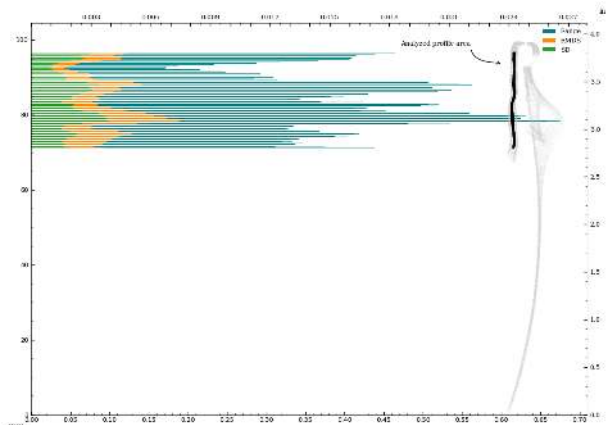
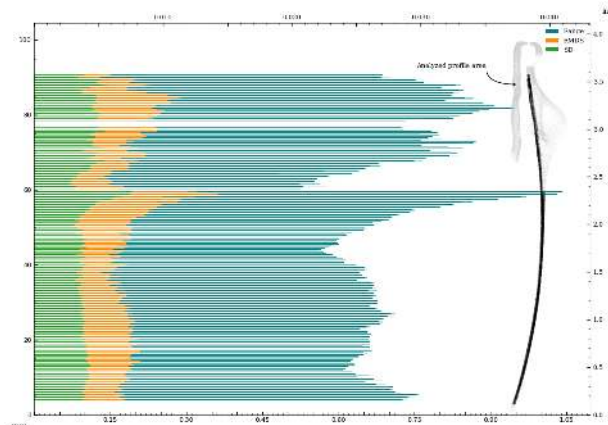
Selected measurements shown below, see full artifact report for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0389	0.197	0.111	0.083	1.233	-0.491	0.742	77140
Interior	0.3736	0.611	0.277	0.212	2.343	-1.082	1.260	73943
Interior separate	0.0115	0.107	0.066	0.042	0.714	-0.347	0.367	74234
	in ²	in	in	in	in	in	in	
Exterior	0.000060	0.0078	0.0044	0.0033	0.0486	-0.0193	0.0292	77140
Interior	0.000579	0.0241	0.0109	0.0083	0.0922	-0.0426	0.0496	73943
Interior separate	0.000018	0.0042	0.0026	0.0016	0.0281	-0.0136	0.0145	74234

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.677	0.481	1.044	0.102	0.068	0.195	0.188	0.121	0.361	427	0.200
Interior	0.406	0.164	0.676	0.053	0.023	0.101	0.100	0.041	0.190	129	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0266	0.0189	0.0411	0.0040	0.0027	0.0077	0.0074	0.0048	0.0142	427	0.0079
Interior	0.0160	0.0064	0.0266	0.0021	0.0009	0.0040	0.0039	0.0016	0.0075	129	0.0079



MV031



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Artifact Precision Report, version 01.20

Material	Basalt
Description	Basalt cylinder vase
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 8, Interior 16

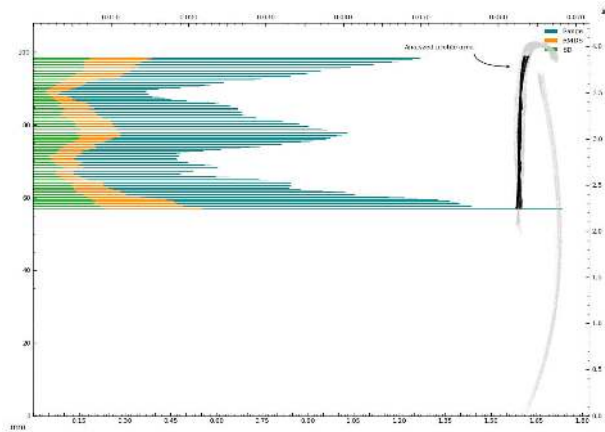
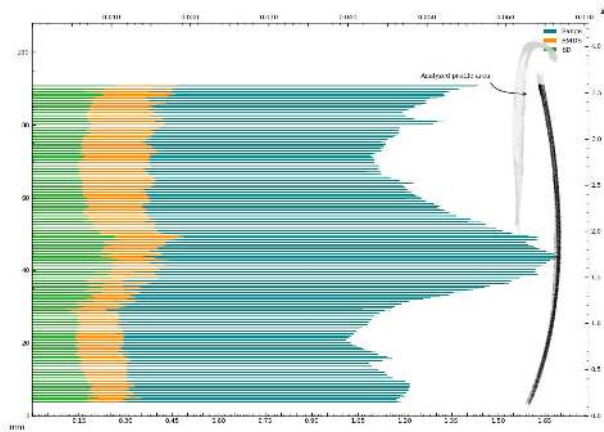
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.1324	0.364	0.192	0.141	2.116	−1.153	0.962	121879
Interior	0.7113	0.843	0.432	0.332	3.277	−1.935	1.342	106258
Interior separate	0.0634	0.252	0.152	0.095	1.535	−0.654	0.881	106042
	in ²	in	in	in	in	in	in	
Exterior	0.000205	0.0143	0.0076	0.0056	0.0833	−0.0454	0.0379	121879
Interior	0.001103	0.0332	0.0170	0.0131	0.1290	−0.0762	0.0528	106258
Interior separate	0.000098	0.0099	0.0060	0.0038	0.0605	−0.0258	0.0347	106042

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.207	1.006	1.705	0.175	0.106	0.303	0.369	0.235	0.488	439	0.200
Interior	0.785	0.356	1.735	0.119	0.038	0.238	0.220	0.074	0.552	210	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0475	0.0396	0.0671	0.0069	0.0042	0.0119	0.0145	0.0093	0.0192	439	0.0079
Interior	0.0309	0.0140	0.0683	0.0047	0.0015	0.0094	0.0086	0.0029	0.0217	210	0.0079



MV032

Artifact
Image
pending



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Artifact Precision Report, version 01.20	
Material	Basalt
Description	Basalt barrel vase, two lug handles, horizontally perforated, and flat base.
Collection	Petrie Museum of Egyptian Archaeology
Provenance	Petrie Museum of Egyptian Archaeology (London), recovered by Flinders Petrie
Precision Score	Exterior 17, Interior 38

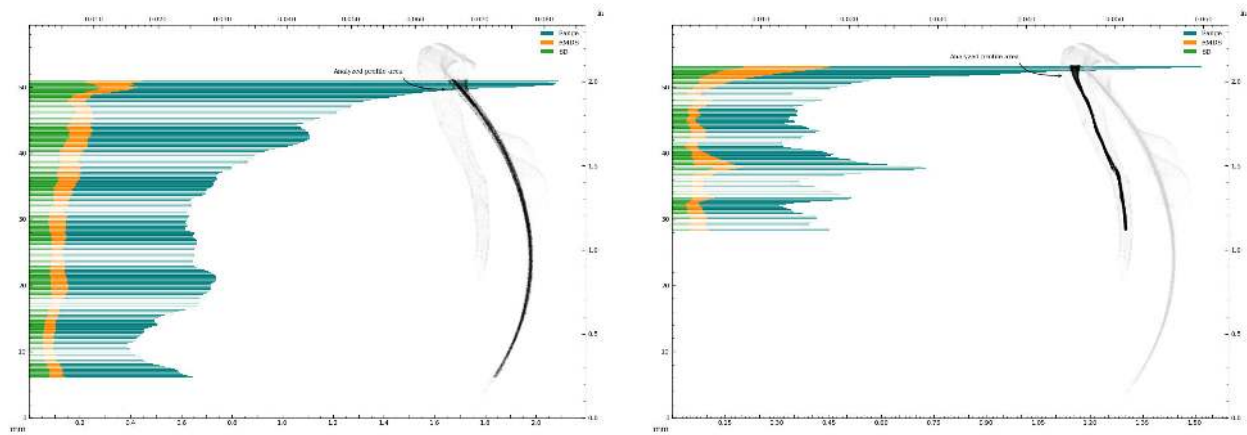
Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.0575	0.240	0.173	0.076	2.613	−1.070	1.543	273044
Interior	0.9676	0.984	0.498	0.356	4.751	−2.722	2.029	120180
Interior separate	0.0260	0.161	0.112	0.049	1.493	−0.782	0.710	118483
	in ²	in	in	in	in	in	in	
Exterior	0.000089	0.0094	0.0068	0.0030	0.1029	−0.0421	0.0607	273044
Interior	0.001500	0.0387	0.0196	0.0140	0.1871	−0.1072	0.0799	120180
Interior separate	0.000040	0.0063	0.0044	0.0019	0.0588	−0.0308	0.0280	118483

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	0.675	0.380	2.090	0.087	0.052	0.285	0.147	0.096	0.447	225	0.200
Interior	0.406	0.305	1.520	0.053	0.037	0.206	0.097	0.066	0.449	127	0.200
	in	in	in	in	in	in	in	in	in		in
Exterior	0.0266	0.0149	0.0823	0.0034	0.0020	0.0112	0.0058	0.0038	0.0176	225	0.0079
Interior	0.0160	0.0120	0.0598	0.0021	0.0014	0.0081	0.0038	0.0026	0.0177	127	0.0079



REF01

Artifact
Image
pending



Artifact Precision Report, version 01.20

Material

Description Zeiss 1.00000 inch reference sphere

Collection

Provenance

Precision Score Exterior 43943

Selected measurements shown below, see [full artifact report](#) for further details.

Surface Variability

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
REF01	0.0000	0.005	0.003	0.002	0.034	−0.017	0.017	11219
	in ²	in	in	in	in	in	in	
REF01	0.000000	0.0002	0.0001	0.0001	0.0013	−0.0007	0.0007	11219

Circularity measurements

Area	Range			Standard Deviation			Median Absolute Deviation			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
REF01	0.022	0.018	0.033	0.003	0.002	0.004	0.005	0.004	0.006	54	0.200
	in	in	in	in	in	in	in	in	in		in
REF01	0.0009	0.0007	0.0013	0.0001	0.0001	0.0001	0.0002	0.0001	0.0002	54	0.0079

